# Learning to Protect Communications with Adversarial Neural Cryptography

*Theoretical Report*

Based on Abadi & Andersen (2016) [1]

Implementation: `main_enhanced.py` — TensorFlow 2.x · Keras


| | | | |
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# Abstract


**Overview**

This report presents a comprehensive theoretical investigation of **adversarial neural cryptography** — a paradigm in which neural networks autonomously learn to encrypt and decrypt communications *without any prescribed cryptographic algorithm*. The work is grounded in the landmark study of Abadi and Andersen [1] and is accompanied by a full TensorFlow 2.x implementation in `main_enhanced.py`, which realises the theoretical system described throughout this report.

The system models a three-party communication scenario involving a sender (Alice), a legitimate receiver (Bob), and a passive adversary (Eve). Alice and Bob share a secret key $K \in \{-1, +1\}^{K_{\text{size}}}$ and are jointly optimised — via stochastic gradient descent using the Adam optimiser [4] — to minimise Bob's message reconstruction error while simultaneously maximising Eve's reconstruction error. Eve is trained competitively and independently, receiving a configurable number of training steps per round to approximate the strongest possible eavesdropping attack.

The core network architecture — termed **"mix & transform"** by Abadi and Andersen — consists of a fully-connected (FC) layer that blends the plaintext and key representations, followed by four one-dimensional convolutional layers, all with stride=1 and SAME padding, and sigmoid/tanh activations. This design allows the network to learn locality and bit-mixing as *emergent properties*, rather than encoding them by construction. All three agents share this architecture; Eve is structurally disadvantaged by receiving only the ciphertext as input, with no access to the shared key.

The loss functions are carefully designed around an important insight: the target for Eve is *indistinguishability from a random guess*, not maximum error. For $N$-bit messages, random guessing yields $N/2$ bits wrong on average. The quadratic adversarial penalty:

$$\mathcal{L}_{\text{adv}} = \frac{\left(\frac{N}{2} - \|\mathbf{p} - \hat{\mathbf{p}}_E\|_1\right)^2}{\left(\frac{N}{2}\right)^2} \tag{1}$$

encodes this goal smoothly, placing strong gradient pressure on Alice and Bob when Eve is close to the random-guessing threshold.

The implementation, `main_enhanced.py`, is built on TensorFlow 2.x using Keras model subclassing and eager execution with `@tf.function` compilation. It supports model checkpointing and reloading, post-training Eve robustness evaluation via multiple reset-and-retrain cycles, real ASCII text encryption via block-based chunking, structured image-patch encryption using real image data, a configurable training step ratio between Alice/Bob and Eve, three loss function variants (quadratic, linear, binary cross-entropy), an optional self-attention layer in the architecture, and automated loss-curve plotting.

Experimental results for $N = 16$-bit plaintexts and keys demonstrate that the system achieves its security objectives in approximately 70% of training runs, with Bob's reconstruction error converging near zero and Eve's error stabilising near the random-guessing baseline of 8 bits. This report provides a rigorous theoretical analysis covering system architecture, loss design, training dynamics, security properties, and a detailed walkthrough of the implementation in `main_enhanced.py`.

**Keywords**

Neural cryptography · Adversarial training · Generative adversarial networks · Symmetric encryption · Alice–Bob–Eve framework · Convolutional neural networks · Self-attention · TensorFlow 2 · ASCII bitwise encoding · Image patch encryption · Loss function variants

# 1. Introduction

## 1.1 Background and Motivation

The protection of digital communications has been a foundational challenge since the emergence of computer networking. Classical cryptographic systems — including the Advanced Encryption Standard (AES) [7], RSA [8], and elliptic-curve cryptography — have served as the bedrock of secure data transmission for decades. Their security derives from well-understood mathematical hardness assumptions: the intractability of integer factorisation, the difficulty of computing discrete logarithms, and the one-wayness of carefully designed permutation functions. While these approaches are supported by rigorous formal proofs under clearly stated computational models, they are inherently *static*: their algorithms are fixed by human design, their resistance to attack depends on the long-term validity of number-theoretic conjectures, and their adaptation to novel threat landscapes requires substantial expert intervention.

In parallel, the field of machine learning has undergone a profound transformation. The advent of deep neural networks [6] has demonstrated that highly complex tasks — long believed to require explicit algorithmic specification — can be learned end-to-end from data and gradient-based feedback alone. This paradigm shift places two powerful disciplines in productive tension, and prompts a question of both theoretical and practical significance:

**Central Research Question**

*Can neural networks autonomously learn to communicate securely — developing their own encryption and decryption strategies — without being taught any cryptographic primitive, using only an adversarial training objective?*

The landmark work of Abadi and Andersen [1], *"Learning to Protect Communications with Neural Cryptography,"* provided the first compelling affirmative answer. By framing secure communication as an adversarial game between three neural network agents — Alice (sender), Bob (receiver), and Eve (passive eavesdropper) — they demonstrated that networks could spontaneously develop encryption-like behaviour through end-to-end gradient optimisation alone, without any pre-specified cryptographic algorithm. This finding opened an entirely new research frontier: **adversarial neural cryptography**.

The motivation for studying such systems extends well beyond academic curiosity. As artificial intelligence becomes increasingly embedded in communication infrastructure, adaptive and automatically discovered cryptographic mechanisms carry significant practical appeal. Furthermore, studying what neural networks learn when tasked with secrecy deepens our understanding of information hiding, representation learning, and the fundamental limits of optimisation in multi-agent settings. Notably, the original paper also observes a fundamental incompatibility: classical cryptographic functions are generally not differentiable, making them difficult to integrate into gradient-based training pipelines. Neural cryptography sidesteps this by *learning* the cryptographic function from scratch.

## 1.2 Problem Statement

The core problem addressed in this work can be stated precisely as follows. A sender (Alice) and a legitimate receiver (Bob) share a secret key $K$ and wish to communicate a plaintext message $P$

over a channel that is also observed by a passive adversary (Eve). The question is whether two neural networks representing Alice and Bob can co-develop a shared communication protocol that is intelligible to Bob but unintelligible to Eve, using only the shared key and gradient descent, and without any pre-specified notion of how to encrypt or decrypt.

***Formal Setup***

Let $P \in \{-1, +1\}^N$ denote the plaintext and $K \in \{-1, +1\}^{K_{\text{size}}}$ the shared secret key. The three agents are parameterised as follows:

- **Alice** computes ciphertext $C = f_{\theta_A}(P, K)$, where $\theta_A$ are her learned weights.
- **Bob** reconstructs the plaintext as $\hat{P}_B = g_{\theta_B}(C, K)$, with access to the shared key.
- **Eve** attempts to recover the plaintext as $\hat{P}_E = h_{\theta_E}(C)$, *with no access to* $K$.

The distance function used throughout is the $\ell_1$ norm:

$$d(P,\ P') \ = \ \sum_{i=1}^{N} \left| P_i - P'_i \right| \tag{2}$$

The per-example losses are defined as:

$$L_{\text{E}}(\theta_A, \theta_E,\ P, K) = d(P,\ h_{\theta_E}(f_{\theta_A}(P, K))) \tag{3}$$

$$L_{\text{B}}(\theta_A, \theta_B,\ P, K) = d(P,\ g_{\theta_B}(f_{\theta_A}(P, K),\ K)) \tag{4}$$

$$L_{\text{AB}}(\theta_A, \theta_B) = L_{\text{B}}(\theta_A, \theta_B) \ - \ L_{\text{E}}(\theta_A,\ \text{OE}(\theta_A)) \tag{5}$$

where $\text{OE}(\theta_A) = \arg\min_{\theta_E} L_{\text{E}}(\theta_A, \theta_E)$ denotes the *optimal Eve* for a given Alice. The training objectives are:

$$(\text{Alice \& Bob}): \quad (\theta_A^*, \theta_B^*) = \arg\min_{\theta_A, \theta_B} \ L_{\text{AB}}(\theta_A, \theta_B) \tag{6}$$

$$(\text{Eve}): \quad \theta_E^* = \arg\min_{\theta_E} \ L_{\text{E}}(\theta_A, \theta_E) \tag{7}$$

The security target for Alice and Bob is to drive Eve's error to the random-guessing baseline of $N/2$ bits wrong — the error an adversary would achieve by outputting uniformly random binary predictions, regardless of the ciphertext.

This problem differs fundamentally from classical cryptography in three key respects:

1. **No pre-specified algorithm.** The encryption and decryption functions are entirely learned through optimisation.
2. **No formal proof of security.** Resistance to eavesdropping is evaluated empirically through competitive adversarial training, not mathematical reduction.
3. **No domain-specific knowledge.** The networks must bootstrap their own cipher purely from the training signal.

## 1.3 Research Objectives and Scope

This report provides a rigorous theoretical treatment of adversarial neural cryptography, anchored in the Abadi and Andersen [1] paper and realised through the TensorFlow 2 implementation `main_enhanced.py`. The specific objectives are:

**Research Objectives**

**O1. Theoretical Foundations.** Survey the foundational concepts of classical cryptography, information theory, and deep learning that underpin the security goals and architectural choices of neural cryptographic systems.

**O2. Architecture Analysis.** Provide a detailed exposition of the mix & transform architecture, explaining how the interplay of FC and convolutional layers enables emergent key-dependent encryption, and how the optional self-attention layer extends this capacity.

**O3. Loss Function Design.** Formally derive and analyse the loss functions used for Alice, Bob, and Eve, explaining the rationale for the quadratic adversarial penalty and comparing it against linear and binary cross-entropy variants.

**O4. Training Dynamics.** Analyse the adversarial training process, including its non-monotonic convergence behaviour, the saddle-point nature of the optimisation, and the impact of the configurable training step ratio.

**O5. Implementation Study.** Provide a thorough annotated walkthrough of `main_enhanced.py`, mapping each design decision directly to its theoretical justification.

**O6. Security Evaluation.** Critically assess the cryptographic security properties of the learned ciphers through post-training Eve retraining, contrasting empirical robustness with formal definitions such as IND-CPA and semantic security.

The scope of this report is primarily theoretical and analytical. While experimental results are referenced to substantiate theoretical claims, the primary contribution lies in building a coherent conceptual framework that bridges machine learning theory and cryptographic science. The report does not address asymmetric (public-key) neural cryptography or selective protection in depth, though both are discussed briefly in the context of future research directions.

## 1.4 Overview of Contributions

The principal contributions of this report are threefold.

**Contribution 1: Unified Theoretical Framework.** This report synthesises insights from cryptography, information theory, and deep learning into a single coherent framework for understanding adversarial neural cryptography. Such a synthesis is largely absent from the existing literature, which tends to treat these disciplines in isolation. In particular, the report bridges Shannon's (1949) model of perfect secrecy [2] with the adversarial GAN-inspired training framework of Goodfellow et al. [3], contextualising both within the specific architecture and loss design of the Abadi and Andersen system.

**Contribution 2: Code-Grounded Analysis.** Unlike purely theoretical treatments, this report grounds its analysis directly in `main_enhanced.py`. Each theoretical concept is mapped to a concrete code artefact: the loss functions `alice_bob_loss()` and `eve_loss()` are formally derived; the `CipherNet` and `AttackerNet` Keras models are annotated layer by layer; and the full training loop — including model checkpointing, multiple Eve retraining, configurable step ratios, loss function variants, self-attention, message complexity modes, and automated plotting — is analysed step by step with theoretical justification at each stage.

**Contribution 3: Critical Security Assessment.** The report provides a formal security analysis that goes beyond empirical evaluation to examine whether learned ciphers satisfy formal cryptographic definitions. It identifies the precise gap between *empirical robustness* — demonstrated by Eve's reconstruction error converging to the random-guessing baseline across multiple post-training retraining cycles — and *provable security*, which requires guarantees against all polynomial-time adversaries rather than a single trained neural network.

## 1.5 Report Organisation

The remainder of this report is organised as follows. Section 2 establishes the theoretical background, covering classical cryptographic primitives, Shannon's model of information-theoretic secrecy, foundational neural network concepts, the generative adversarial framework, the Adam optimiser, and the self-attention mechanism. Section 3 provides a detailed exposition of the system architecture, including the Alice–Bob–Eve framework, the mix & transform network design, and the attention-enhanced variant. Section 4 formally analyses the loss functions and optimisation objectives, including the three loss function variants introduced in the enhanced implementation. Section 5 examines training dynamics, including convergence behaviour and the saddle-point nature of the adversarial game. Section 6 covers message complexity extensions: random binary, ASCII text, and image-patch modes. Section 7 presents the implementation details with a full annotated walkthrough of `main_enhanced.py`. Section 8 evaluates the security properties of the learned cryptosystem, including post-training Eve robustness evaluation. Section 9 presents and discusses experimental results across all configurations: baseline random-binary mode, loss function variants, training step ratios, the self-attention mechanism, post-training robustness, ASCII text mode under both loss variants, and structured image-patch training on CIFAR-10 data. Section 10 provides a critical discussion of the approach's strengths, limitations, and open problems. Section 11 concludes the report.

# 2. Theoretical Background

This section establishes the theoretical foundations required to understand adversarial neural cryptography. We cover classical cryptographic principles, Shannon's information-theoretic model of secrecy, the neural network components used in the system, the generative adversarial training framework, the Adam optimiser, and the self-attention mechanism. Each subsection is directly connected to a design decision in `main_enhanced.py`.

## 2.1 Classical Cryptography Fundamentals

***The Symmetric Encryption Model***

Classical cryptography is broadly concerned with algorithms and protocols that ensure the *secrecy* and *integrity* of information. The present work focuses exclusively on **symmetric encryption** — also called *shared-key* encryption — in which both the sender and the legitimate receiver possess the same secret key, which is kept hidden from all other parties.

Formally, a symmetric encryption scheme is a triple of efficient algorithms ($\mathsf{Gen}$, $\mathsf{Enc}$, $\mathsf{Dec}$) satisfying:

$$K \leftarrow \mathsf{Gen}(1^\lambda) \qquad \text{(key generation)}$$

$$C \leftarrow \mathsf{Enc}(K,\, P) \qquad \text{(encryption)}$$

$$P = \mathsf{Dec}(K,\, C) \qquad \text{(decryption)}$$

where $\lambda$ is the security parameter, $P$ is the plaintext, $C$ is the ciphertext, and correctness requires that $\mathsf{Dec}(K, \mathsf{Enc}(K, P)) = P$ for all valid $P$ and $K$.

In the neural cryptography setting of `main_enhanced.py`, these roles are fulfilled by learned functions: Alice's network implements $\mathsf{Enc}$ and Bob's network implements $\mathsf{Dec}$. The key $K$ is generated uniformly at random from $\{-1, +1\}^{K_{\text{size}}}$ for each plaintext:

```
key = tf.cast(
    2 * tf.random.uniform([batch_size, key_size], 0, 2, dtype=tf.int32)
        - 1,
    tf.float32
)   # K in {-1, +1}^key_size
```

Listing 1: Key and plaintext generation in `main_enhanced.py`

***Confidentiality vs. Integrity***

Two fundamental security properties are distinguished in cryptography:

- **Confidentiality** (secrecy): an adversary who observes the ciphertext $C$ learns nothing about the plaintext $P$. This is the property targeted by adversarial neural cryptography.
- **Integrity** (authenticity): an adversary cannot modify the ciphertext without detection. Neural cryptography as studied here does *not* address integrity — Eve is a passive adversary who only observes, never modifies.

The threat model used throughout this report is the **ciphertext-only attack (COA)**: Eve observes ciphertexts $C$ but has no access to corresponding plaintexts or the key. This is the weakest standard attack model, and the one assumed in [1]. The security goal is for Eve to perform no better than *random guessing* on the plaintext bits.

#### *Indistinguishability (IND-CPA)*

Modern cryptography formalises confidentiality through the notion of **indistinguishability under chosen-plaintext attack (IND-CPA)**. Informally, a scheme is IND-CPA secure if no efficient adversary can distinguish between ciphertexts of two chosen plaintexts with probability significantly greater than $\frac{1}{2}$ [5]. Neural cryptography does not achieve IND-CPA in the formal sense — there is no proof against all polynomial-time adversaries — but it pursues the weaker empirical goal of defeating a specific trained neural network adversary. This gap is discussed in detail in Section 8.

## 2.2 Shannon's Theory of Secrecy

#### *Information-Theoretic Secrecy*

Claude Shannon's 1949 paper [2] established the mathematical foundations of secret communication. Shannon defined **perfect secrecy** as the condition under which observing the ciphertext $C$ provides *zero information* about the plaintext $P$, regardless of the adversary's computational power:

$$H(P \mid C) \;=\; H(P) \tag{8}$$

where $H(\cdot)$ denotes the Shannon entropy and $H(P \mid C)$ is the conditional entropy of the plaintext given the ciphertext. Equivalently, $P$ and $C$ must be statistically independent: $\Pr[P = p \mid C = c] = \Pr[P = p]$ for all $p, c$.

#### *Entropy*

The **Shannon entropy** of a discrete random variable $X$ taking values in $\mathcal{X}$ is:

$$H(X) \;=\; -\sum_{x \in \mathcal{X}} \Pr[X = x] \log_2 \Pr[X = x] \tag{9}$$

For a uniformly distributed $N$-bit binary string, $H(X) = N$ bits — the maximum possible entropy. Both the plaintext $P$ and key $K$ in our system are sampled uniformly from $\{-1, +1\}^N$, so they carry maximum entropy. This is the ideal theoretical baseline: Eve begins with no information advantage beyond knowledge of the distribution.

#### *The One-Time Pad as the Theoretical Ideal*

Shannon proved that the **one-time pad (OTP)** achieves perfect secrecy when the key is uniformly random, as long as the key is at least as long as the plaintext and is never reused. For binary strings, OTP encryption is simply bitwise XOR:

$$C_i = P_i \oplus K_i, \qquad P_i = C_i \oplus K_i \tag{10}$$

The OTP serves as the theoretical ideal against which neural cryptography is implicitly compared. The mix & transform architecture is deliberately designed so that XOR *could* emerge as a solution — the FC layer can represent XOR and the convolutional layers can preserve it — but the network is not constrained to use it. Empirically, Abadi and Andersen observe that the learned cipher is *not* pure XOR: single-bit changes in $K$ or $P$ propagate to multiple ciphertext positions, a property reminiscent of *diffusion* in classical ciphers.

#### *Why $N/2$ is the Random-Guessing Baseline*

A central quantity in this system is the **random-guessing baseline** of $N/2$ bits wrong. This arises naturally from information theory. If EVE learns nothing about $P$ from $C$ — that is, if perfect secrecy holds — then her optimal strategy is to guess each bit independently and uniformly. For each of the $N$ bits:

$$\Pr[\hat{P}_{E,i} \neq P_i] = \tfrac{1}{2} \tag{11}$$

By linearity of expectation, the expected number of wrong bits is:

$$\mathbb{E}\left[\sum_{i=1}^{N} \mathbf{1}[\hat{P}_{E,i} \neq P_i]\right] = \frac{N}{2} \tag{12}$$

For $N = 16$, this gives the target of exactly 8 bits wrong. The adversarial penalty in `main_enhanced.py` is specifically designed to push EVE's error toward this value — not above it (which would be trivially reversible) and not below it (which would indicate information leakage). This is formalised in Section 4.

## 2.3 Neural Network Foundations

The mix & transform architecture used by ALICE, BOB, and EVE is composed of three fundamental building blocks: fully-connected layers, one-dimensional convolutional layers, and nonlinear activation functions. We describe each in turn, directly connecting them to the Keras implementation in `main_enhanced.py`.

#### *Fully-Connected (Dense) Layers*

A **fully-connected** (FC) layer, also called a *Dense* layer, computes an affine transformation of its input:

$$\mathbf{y} = W\mathbf{x} \tag{13}$$

where $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, and $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$. Because each output neuron receives contributions from *all* input neurons, the FC layer enables **global mixing** of its input. In our system, the first FC layer of ALICE's network receives the concatenation $[P \,\|\, K]$ of size $N + K_{\text{size}} = 32$ (for the default $N = 16$) and mixes all plaintext and key bits together:

```
self.fc = keras.layers.Dense(N, use_bias=False)
# Input: [P || K] of size N + key_size = 32
# Output: mixed representation of size N = 16
```

Listing 2: FC layer in `CipherNet` — global key-plaintext mixing

No bias is used, consistent with the original architecture. The absence of bias means the output is a *pure linear combination* of the inputs — the nonlinearity is deferred to subsequent convolutional layers.

#### *One-Dimensional Convolutional Layers*

A **1D convolutional layer** slides a bank of $F$ learnable filters of width $w$ over a sequence of length $L$ with $C_{\text{in}}$ channels, producing an output of $C_{\text{out}}$ channels:

$$y_j[t] = \sum_{c=1}^{C_{\text{in}}} \sum_{k=0}^{w-1} W_j[c,k]\, x_c[t \cdot s + k] \tag{14}$$

where $s$ is the *stride* and $j = 1, \ldots, C_{\text{out}}$. The four convolutional layers in `CipherNet` are configured as shown in Table 1.

Table 1: Convolutional layer configuration in `CipherNet` and `AttackerNet`. "Depth" refers to (input channels, output channels). Activation is sigmoid except the final layer which uses tanh.

| **Layer** | **Window** | **Depth** | **Stride** | **Padding** | **Activation** |
|---|---|---|---|---|---|
| Conv1 | 4 | $1 \to 2$ | 1 | SAME | Sigmoid |
| Conv2 | 2 | $2 \to 4$ | 1 | SAME | Sigmoid |
| Conv3 | 1 | $4 \to 4$ | 1 | SAME | Sigmoid |
| Conv4 | 1 | $4 \to 1$ | 1 | SAME | **Tanh** |

With stride=1 and SAME padding throughout, the sequence length remains $N = 16$ at every convolutional layer. Conv2 expands the channel depth from 2 to 4 while preserving length. Conv3 and Conv4 further refine the representation. The final `tf.squeeze` removes the trailing channel dimension, recovering the $N$-dimensional output.

The key design insight is that convolutions learn **local** combinations of nearby bits — but *which* bits are combined, and how, is entirely learned. This contrasts with XOR (which pairs bits at identical positions) or a second FC layer (which would re-mix globally and be harder to train). The ordering of FC then convolutions was specifically chosen so that global mixing happens first, and local refinement follows.

***Activation Functions: Sigmoid vs. Tanh***

Two activation functions are used in the architecture:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1) \tag{15}$$

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}} \in (-1, 1) \tag{16}$$

Sigmoid is applied after Conv1, Conv2, and Conv3. Its output in $(0, 1)$ allows the intermediate representations to encode soft binary-like decisions. **Tanh** is applied exclusively after the final layer, Conv4. This is critical: since the plaintext and key inputs are in $\{-1, +1\}$, the output of Alice (the ciphertext) and Bob/Eve (the decrypted message) must also lie in $[-1, +1]$ to be comparable with the input. Tanh enforces this range constraint at the output.

***Backpropagation and Gradient-Based Learning***

All three networks are trained by **backpropagation** — the application of the chain rule to compute gradients of the scalar loss with respect to all trainable parameters:

$$\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}^{(\ell)}} \cdot \frac{\partial \mathbf{a}^{(\ell)}}{\partial W^{(\ell)}} \tag{17}$$

where $\mathbf{a}^{(\ell)}$ is the pre-activation output of layer $\ell$. In `main_enhanced.py`, TensorFlow 2's automatic differentiation engine handles this via `tf.GradientTape`:

```python
with tf.GradientTape() as tape:
    cipher  = alice(tf.concat([o_msg, key], axis=1))
    dec_bob = bob(tf.concat([cipher, key], axis=1))
    dec_eve = eve(cipher)
    loss    = alice_bob_loss(o_msg, dec_bob, dec_eve, msg_size, variant
        )
grads = tape.gradient(loss, alice.trainable_variables
                            + bob.trainable_variables)
optimizer_ab.apply_gradients(zip(grads, alice.trainable_variables
                                        + bob.trainable_variables))
```

Listing 3: Gradient computation using `tf.GradientTape`

## 2.4 Generative Adversarial Networks

### *The GAN Framework*

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [3], provide the conceptual framework for adversarial neural cryptography. A GAN consists of two networks trained in opposition:

- A **Generator** $G$ that produces outputs intended to fool the discriminator.
- A **Discriminator** $D$ that attempts to distinguish generated outputs from real data.

Their training is formalised as a **minimax game**:

$$\min_G \max_D \;\; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] \;+\; \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))] \tag{18}$$

Adversarial neural cryptography adapts this framework: Alice (and Bob) play the role of the generator, attempting to produce ciphertexts that are uninformative to Eve; Eve plays the role of the discriminator, attempting to extract information from those ciphertexts. The key difference from standard GANs is that the adversarial goal is not image realism but *information concealment*: Eve must be unable to reconstruct the plaintext, not merely unable to distinguish ciphertexts from random noise.

### *Nash Equilibrium*

The theoretical solution concept for a minimax game is the **Nash equilibrium** — a pair of strategies $(\theta^*_{AB}, \theta^*_E)$ such that neither party can improve their outcome by unilaterally changing their strategy:

$$\mathcal{L}_{AB}(\theta^*_{AB},\, \theta^*_E) \;\leq\; \mathcal{L}_{AB}(\theta_{AB},\, \theta^*_E) \quad \forall\, \theta_{AB} \tag{19}$$

$$\mathcal{L}_E(\theta^*_{AB},\, \theta^*_E) \;\leq\; \mathcal{L}_E(\theta^*_{AB},\, \theta_E) \quad \forall\, \theta_E \tag{20}$$

At equilibrium, Eve achieves the random-guessing error of $N/2$ bits wrong — she cannot do better regardless of her strategy — while Alice and Bob achieve perfect reconstruction. Whether gradient-based training converges to this equilibrium is not guaranteed in general.

### *Why Adversarial Training Does Not Converge Monotonically*

Standard supervised learning minimises a fixed loss and converges monotonically (under mild conditions). Adversarial training does not, because the loss landscape for each player *changes* as the other player's parameters are updated. This produces the characteristic non-monotonic training curves observed in both GANs and neural cryptography:

1. **Phase 1 — Random exploration.** Both Bob and Eve perform poorly. The ciphertext carries little structure.

2. **Phase 2 — Trivial cipher discovery.** Alice and Bob find a simple communication protocol (e.g., akin to rot-13). Eve begins to exploit it.

3. **Phase 3 — Eve breaks the trivial cipher.** Eve's error drops; Alice and Bob's loss increases.

4. **Phase 4 — Key exploitation.** Under pressure from Eve, Alice and Bob discover key-dependent encryption. Eve's error rises back toward $N/2$.

This four-phase dynamic is directly visible in the loss curves produced by `save_loss_curves()` in `main_enhanced.py` and mirrors the training graph described in Section 2.5 of [1]. Unlike typical neural network training, *monotonicity in the number of steps is not expected* — the system exhibits dynamics more reminiscent of co-evolutionary processes than gradient descent on a fixed landscape.

## 2.5 The Adam Optimiser

### *Adaptive Moment Estimation*

All three networks — Alice, Bob, and Eve — are trained using the **Adam** (Adaptive Moment Estimation) optimiser [4]. Adam maintains exponentially decaying estimates of the first moment (mean) and second moment (uncentred variance) of the gradients:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\, g_t \tag{21}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\, g_t^2 \tag{22}$$

After bias correction, the parameter update is:

$$\theta_t \;=\; \theta_{t-1} \;-\; \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} \tag{23}$$

where $\hat{m}_t = m_t/(1-\beta_1^t)$, $\hat{v}_t = v_t/(1-\beta_2^t)$ are the bias-corrected estimates, $\alpha$ is the learning rate, and $\varepsilon$ is a small constant for numerical stability. Default values $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ are used throughout.

```
opt_ab  = keras.optimizers.Adam(args.lr)    # for Alice + Bob
opt_eve = keras.optimizers.Adam(args.lr)    # for Eve (separate)
# Default: args.lr = 0.0008
```

Listing 4: Adam optimisers in `main_enhanced.py`

### *Why a Learning Rate of* 0.0008

The learning rate of $\alpha = 0.0008$ is taken directly from the original experimental setup of [1]. It represents a carefully chosen compromise:

- **Too large** ($\alpha \gg 0.001$): training becomes unstable. In the adversarial setting, large updates from one player can destabilise the other, leading to divergence or persistent oscillation.

- **Too small** ($\alpha \ll 0.0001$): learning is too slow. Given the non-monotonic training dynamics, Alice and Bob need sufficient step size to respond meaningfully to Eve's adaptations within the training budget of 20,000 steps.

- $\alpha = 0.0008$ sits just below the standard default of 0.001, providing stable but responsive training in the adversarial regime.

***Why the Learning Rate is NOT Decayed***

In standard supervised learning, it is common practice to reduce the learning rate over time (e.g., cosine annealing or step decay) to allow fine-grained convergence near the optimum. This report's system deliberately **does not decay the learning rate**, for the following reason stated explicitly in [1]:

**Design Rationale: No Learning Rate Decay**

ALICE, BOB, and EVE must remain capable of *responding strongly to changes in other components* throughout training. A decayed learning rate would reduce this responsiveness, potentially causing ALICE and BOB to fail to adapt when EVE breaks their cipher in the later stages of training, or causing EVE to stop exploiting newly discoverable weaknesses as ALICE's encryption evolves.

This is a direct consequence of the adversarial training dynamic: unlike a fixed loss landscape, each player's optimal response changes continuously as the others learn. Keeping the learning rate constant ensures all three agents remain *alive* — capable of meaningful adaptation — for the entire training duration.

## 2.6 Self-Attention Mechanism

***Motivation: Beyond Fixed Convolutional Locality***

The standard mix & transform architecture uses convolutional layers with fixed kernel sizes (4, 2, 1, 1) to capture local relationships between sequence positions. While effective, this approach has a fundamental limitation: the *receptive field* of each output position — the set of input positions it can directly combine — is determined by the kernel size, not learned from data. A kernel of size 4 in Conv1 can only directly see 4 adjacent positions; deeper layers can aggregate over larger ranges, but the locality structure is implicit and fixed.

Neural cryptography requires learning *which specific bit positions* to combine in order to generate secure ciphertexts. The convolutional architecture addresses this by learning *how* to combine nearby positions, but the *which* (i.e., the pairing structure) is constrained by spatial proximity. Self-attention removes this constraint: every position can attend to every other position, and the attention pattern — which bits to combine and with what weights — is learned entirely from the training signal.

***Scaled Dot-Product Attention***

The **self-attention** mechanism [11] maps an input sequence $\mathbf{X} \in \mathbb{R}^{L \times d}$ to three projections — **queries** $Q$, **keys** $K$, and **values** $V$ — via learned weight matrices:

$$Q = \mathbf{X}W_Q, \quad K = \mathbf{X}W_K, \quad V = \mathbf{X}W_V \tag{24}$$

The attention output is then:

$$\text{Attention}(Q, K, V) \;=\; \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{25}$$

where $d_k$ is the dimension of the key vectors and $\sqrt{d_k}$ is a scaling factor that prevents the dot products from growing too large, which would push the softmax into saturation regions with near-zero gradients.

The matrix $QK^\top \in \mathbb{R}^{L\times L}$ is the **attention score matrix**: entry $(i, j)$ measures how much position $i$ should attend to position $j$. After softmax normalisation, this becomes the **attention weight matrix**, and the output at each position is a weighted sum of all value vectors.

### *Implementation in `SelfAttention1D`*

In `main_enhanced.py`, the `SelfAttention1D` class implements single-head self-attention over the 1D sequence produced by Conv2. Since all conv layers use stride=1 with SAME padding, this sequence has shape $(B, N, 4) = (B, 16, 4)$ — full length $N$, not $N/2$:

```python
class SelfAttention1D(keras.layers.Layer):
    def __init__(self, units, **kwargs):
        super().__init__(**kwargs)
        self.Wq = keras.layers.Dense(units, use_bias=False)
        self.Wk = keras.layers.Dense(units, use_bias=False)
        self.Wv = keras.layers.Dense(units, use_bias=False)
        self.proj = keras.layers.Dense(units, use_bias=False)

    def call(self, x):
        Q = self.Wq(x)                              # (B, L, units)
        K = self.Wk(x)
        V = self.Wv(x)
        scale   = tf.sqrt(tf.cast(self.units, tf.float32))
        scores  = tf.matmul(Q, K, transpose_b=True) / scale
        weights = tf.nn.softmax(scores, axis=-1)    # (B, L, L)
        context = tf.matmul(weights, V)             # (B, L, units)
        return self.proj(context)
```

Listing 5: `SelfAttention1D` in `main_enhanced.py`

The attention layer is inserted between Conv2 and Conv3, with a **residual connection** and **Layer Normalisation**:

```python
if self.use_attention:
    attn_out = self.attention(net)
    net = self.attn_norm(net + attn_out)  # residual + LayerNorm
```

Listing 6: Residual attention block in `CipherNet.call()`

### *Why a Residual Connection and LayerNorm*

Two stabilisation techniques are applied around the attention layer:

- **Residual connection** (`net + attn_out`): allows the gradient to flow directly through the attention block without passing through the attention weights. This prevents vanishing gradients and allows the network to smoothly interpolate between "use attention" and "ignore attention" as needed. If the attention weights contribute noise early in training, the residual path ensures the convolutional features are still preserved.

- **Layer Normalisation**: normalises the summed output across the feature dimension, reducing internal covariate shift and stabilising training without requiring large batch sizes (unlike Batch Normalisation). This is especially important in the adversarial setting, where loss landscapes shift rapidly as the three agents evolve.

### *Expected Effect on Cryptographic Quality*

The self-attention mechanism is expected to benefit the system in two ways. First, it enables ALICE and BOB to learn **long-range bit dependencies** — for instance, XOR-like interactions between positions that are far apart in the sequence — which the convolutional layers cannot represent without many stacked layers. Second, the attention weight matrix provides a form of **learned permutation**: if position $i$ attends strongly to position $j$ and weakly to all others, the effect approximates routing bit $j$'s information to position $i$, a more flexible version of the bit-mixing performed by the initial FC layer. This increased expressivity is expected to produce ciphers with stronger diffusion properties and potentially higher success rates in the post-training robustness evaluation.

Table 2 summarises the architectural difference between the standard and attention-augmented variants.

Table 2: Architecture comparison: standard vs. attention-augmented `CipherNet`. Input size shown for Alice with $N = 16$, $K_{\text{size}} = 16$.

| **Layer** | **Standard** | **Attention-Augmented** |
|---|---|---|
| Input | $[P\|K]$, size 32 | $[P\|K]$, size 32 |
| FC | Dense(16) | Dense(16) |
| Reshape | $(B, 16, 1)$ | $(B, 16, 1)$ |
| Conv1 | $[4, 1\to2]$ s=1 | $[4, 1\to2]$ s=1 |
| Conv2 | $[2, 2\to4]$ s=1 | $[2, 2\to4]$ s=1 |
| *Attention* | — | *SelfAttention1D(4) + Residual + LN* |
| Conv3 | $[1, 4\to4]$ s=1 | $[1, 4\to4]$ s=1 |
| Conv4 | $[1, 4\to1]$ s=1 | $[1, 4\to1]$ s=1 |
| Output | `squeeze`$(B, 16)$ | `squeeze`$(B, 16)$ |

# 3. System Architecture & Design

This section provides a complete technical description of the adversarial neural cryptography system implemented in `main_enhanced.py`. We describe the Alice–Bob–Eve communication framework, the mix & transform network architecture shared by all three agents, the structural asymmetry that disadvantages Eve, the optional attention-augmented variant, and the data representation conventions used throughout.

## 3.1 The Alice–Bob–Eve Framework

***System Overview***

The adversarial neural cryptography system is built around a classic security scenario involving three parties: Alice (the sender), Bob (the legitimate receiver), and Eve (the passive eavesdropper). Figure 1 illustrates the information flow between the three agents.

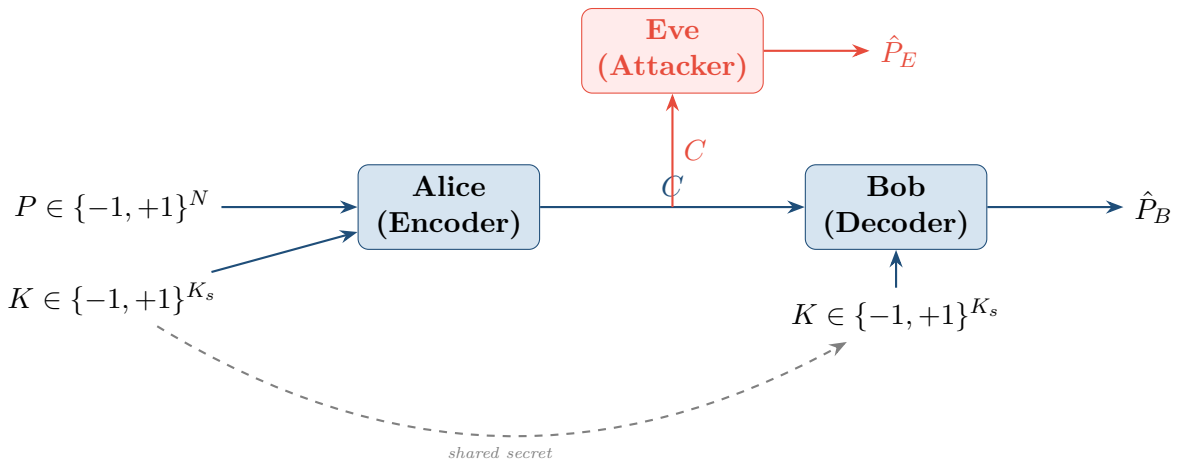


Figure 1: Information flow in the Alice–Bob–Eve framework. Alice encrypts plaintext $P$ using shared key $K$, producing ciphertext $C$. Bob decrypts $C$ using the same key to recover $\hat{P}_B \approx P$. Eve observes $C$ only and attempts to recover $\hat{P}_E \approx P$ without access to $K$.

***Roles and Information Access***

The information available to each agent is summarised in Table 3. The structural asymmetry between Bob (who has the key) and Eve (who does not) is the source of the security advantage: any cipher that effectively uses the key will be harder for Eve to break, since her network must operate without that crucial information.

Table 3: Information access for each agent in the system.

| Agent | Input | Output | Objective | Has Key? |
|---|---|---|---|---|
| ALICE | $P$, $K$ | $C$ (ciphertext) | Encrypt so BOB can decode but EVE cannot | ✓ |
| BOB | $C$, $K$ | $\hat{P}_B$ (decrypted) | Reconstruct $P$ accurately | ✓ |
| EVE | $C$ only | $\hat{P}_E$ (guessed) | Reconstruct $P$ from ciphertext alone | × |

***Threat Model***

The adversary EVE is modelled as a **passive, ciphertext-only attacker**: she observes every ciphertext transmitted by ALICE but cannot intercept, modify, or replay messages, and has no access to the shared key. This is the weakest standard threat model in cryptography. It corresponds directly to the system architecture, where EVE's network receives only the ciphertext tensor as input:

```
cipher  = alice(tf.concat([o_msg, key], axis=1))  # Alice: [P || K]
dec_bob = bob(tf.concat([cipher, key], axis=1))   # Bob:   [C || K]
dec_eve = eve(cipher)                              # Eve:    C only
```

Listing 7: Agent input construction in `train_alice_bob()`

The assumption of one fresh key $K$ per plaintext $P$ means that EVE cannot accumulate statistics across messages encrypted under the same key — each observation is independent. This mimics a one-time-pad key schedule, the strongest possible key usage regime.

## 3.2 The Mix-and-Transform Architecture

***Design Philosophy***

The architecture follows the **mix & transform** design principle introduced in [1]. The four central design goals are:

1. **Expressiveness**: must be capable of representing mixing functions such as XOR, without hard-coding any particular function.

2. **Learned locality**: which bit positions to combine should be a *learned* property, not pre-specified by the architecture.

3. **Key exploitation**: key bits must be able to influence every output bit, enabling key-dependent encryption to emerge naturally.

4. **Output range**: the final output must lie in $[-1, +1]$ to be comparable with the binary input representation.

These goals motivate the specific sequence: one FC layer for global mixing, followed by four 1D convolutional layers for local transformation, with sigmoid activations on all intermediate layers and tanh on the final output layer.

***Complete Architecture Diagram***

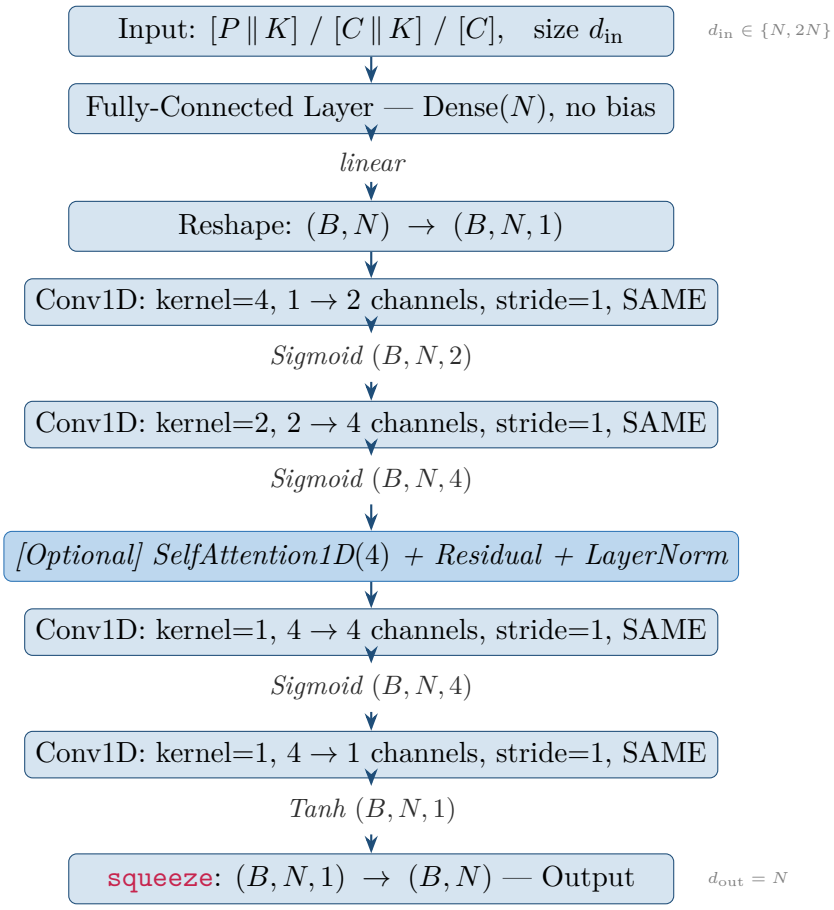


Figure 2: Complete mix & transform architecture for `CipherNet` (Alice and Bob) and `AttackerNet` (Eve). The optional self-attention block is active only when `-attention` is set. Tensor shapes shown for $N = 16$.

## 3.3 Alice's Network: `CipherNet` (Encoder)

Alice's network implements the encryption function $C = f_{\theta_A}(P, K)$. Its input is the concatenation of the plaintext $P \in \{-1, +1\}^N$ and the key $K \in \{-1, +1\}^{K_s}$, giving an input vector of size $N + K_s = 32$ for the default configuration ($N = K_s = 16$).

***FC Layer: Global Mixing***

The FC layer maps the $(N + K_s)$-dimensional input to an $N$-dimensional representation. With no bias, the transformation is purely linear:

$$\mathbf{h}^{(0)} \;=\; W_{\text{fc}}\,[P \,\|\, K]^{\top}, \qquad W_{\text{fc}} \in \mathbb{R}^{N \times (N+K_s)} \tag{26}$$

This layer has $N \times (N + K_s) = 16 \times 32 = 512$ parameters. Because every output neuron receives contributions from all 32 input units, this layer can implement any linear mixing of plaintext and key bits — including pure XOR when $N = K_s$, which corresponds to $W_{\text{fc}}$ being an identity-like matrix that pairs each plaintext bit with its corresponding key bit.

***Convolutional Processing***

After the FC layer the output is reshaped to $(B, N, 1)$ to create a single-channel 1D sequence of length $N$. The four convolutional layers then progressively transform this sequence:

- **Conv1** (kernel=4, 1 → 2 ch, stride=1): slides a width-4 window, capturing short-range interactions between adjacent positions in the mixed representation.
- **Conv2** (kernel=2, 2 → 4 ch, stride=1): expands the channel depth to four while preserving sequence length at $N = 16$. The kernel-2 window captures pairwise interactions between adjacent positions.
- **Conv3** (kernel=1, 4 → 4 ch, stride=1): applies an independent pointwise transformation to each position — equivalent to a per-position FC layer across the channel dimension.
- **Conv4** (kernel=1, 4 → 1 ch, stride=1): reduces to a single channel. The tanh activation constrains outputs to $(-1, +1)$, producing the ciphertext $C$.

***Parameter Count***

Table 4: Parameter count for Alice's `CipherNet` ($N = 16$, $K_s = 16$, no attention).

| Layer | Shape | Parameters | Activation |
|---|---:|---:|---:|
| FC (Dense) | $32 \to 16$ | 512 | Linear |
| Conv1 | $4 \times 1 \times 2 + 2$ | 10 | Sigmoid |
| Conv2 | $2 \times 2 \times 4 + 4$ | 20 | Sigmoid |
| Conv3 | $1 \times 4 \times 4 + 4$ | 20 | Sigmoid |
| Conv4 | $1 \times 4 \times 1 + 1$ | 5 | Tanh |
| **Total** | | **567** | |

Alice's network has only **567 parameters** (512 FC weights plus 55 conv weights and biases) — an intentionally lightweight architecture that ensures emergent encryption arises from genuine learning, not from excessive model capacity. Note that only the FC layer sets `use_bias=False`; all four Conv1D layers retain their default Keras bias terms.

## 3.4 Bob's Network: `CipherNet` (Decoder)

Bob's network implements the decryption function $\hat{P}_B = g_{\theta_B}(C, K)$. It is **architecturally identical** to Alice's network — the same `CipherNet` class is instantiated twice — but its input is the concatenation of the *ciphertext* $C$ and the shared key $K$:

```
dec_bob = bob(tf.concat([cipher, key], axis=1))
# Shape: (B, N + key_size) = (B, 32)
```

Listing 8: Bob's input: ciphertext concatenated with the shared key

***Shared Architecture, Independent Weights***

Although Alice and Bob share the same `CipherNet` architecture, they are **separate Keras model instances** with entirely independent weight tensors $\theta_A \neq \theta_B$. Encryption and decryption need not be the same function. Both $\theta_A$ and $\theta_B$ are updated jointly in each Alice/Bob training step, since the composite loss depends on both:

```
grads = tape.gradient(loss,
    alice.trainable_variables + bob.trainable_variables)
optimizer_ab.apply_gradients(
    zip(grads, alice.trainable_variables + bob.trainable_variables))
```

Listing 9: Joint update of Alice and Bob parameters

This joint optimisation is what enables co-development of a shared protocol: neither can succeed in isolation, and both are updated to maximise their mutual benefit.

## 3.5 Eve's Network: `AttackerNet`

EVE's network implements $\hat{P}_E = h_{\theta_E}(C)$. It uses the `AttackerNet` class, which has **identical internal architecture** to `CipherNet` (FC + 4 convolutions + same activations), but differs in two critical respects.

***Structural Disadvantage: No Key Access***

EVE receives only the ciphertext $C$ as input. Her input size is $N = 16$, not $N + K_s = 32$. Her FC layer is therefore a $16 \times 16 = 256$ parameter matrix, compared to Alice's $32 \times 16 = 512$. The reduced input dimension is the primary source of difficulty: any encryption that effectively uses $K$ creates a family of $2^{K_s}$ possible ciphertexts for each plaintext, and EVE cannot distinguish between them without knowing the key.

Table 5: Structural comparison: `CipherNet` vs. `AttackerNet`.

| **Component** | **CipherNet** (ALICE/BOB) | **AttackerNet** (EVE) |
|---|---|---|
| Input size | $N + K_s = 32$ | $N = 16$ |
| FC layer | $32 \to 16$ | $16 \to 16$ |
| FC parameters | 512 | 256 |
| Conv layers | identical | identical |
| Total params | 567 | 311 |
| Key access | ✓ | × |

***Independent Optimisation***

EVE's parameters $\theta_E$ are updated using a fully separate Adam optimiser instance `opt_eve`. This decoupling is essential: if EVE were updated jointly with ALICE and BOB, no genuine adversarial pressure would exist. The two optimisation problems must remain independent:

```
opt_ab  = keras.optimizers.Adam(args.lr)  # for Alice + Bob
opt_eve = keras.optimizers.Adam(args.lr)  # for Eve -- fully
    independent
```

Listing 10: Separate independent optimisers

## 3.6 Data Representation

***Binary Encoding:*** $\{-1, +1\}$ ***vs.*** $\{0, 1\}$

The system uses $\{-1, +1\}$ rather than $\{0, 1\}$ as the binary representation. In $\{-1, +1\}$ space, the XOR operation corresponds to *multiplication*:

$$a \oplus b \quad \longleftrightarrow \quad a \cdot b, \qquad a, b \in \{-1, +1\} \tag{27}$$

Multiplication is differentiable, making XOR-like interactions expressible by smooth neural network computations. In $\{0, 1\}$ space, XOR has no smooth algebraic analogue, making it harder for gradient descent to discover.

***Floating-Point Ciphertext***

While plaintexts and keys are constrained to $\{-1, +1\}^N$, the ciphertext $C$ is a vector of *real-valued* floating-point numbers in $(-1, +1)$ enforced by the tanh output. This continuous ciphertext space allows gradient information to flow smoothly from Eve's loss back through the ciphertext to Alice's parameters, enabling end-to-end training. Without this continuity, gradient-based optimisation of Alice would not be possible.

# 4. Loss Functions & Optimisation Objectives

The loss functions are the heart of the adversarial neural cryptography system. They encode the security objectives, determine what each agent learns, and govern the competitive dynamics of training. This section formally derives each loss, analyses the design choices behind them, and compares the three variants implemented in `main_enhanced.py`.

## 4.1 The L1 Distance Function

All loss functions in this system are built on the **L1 distance**:

$$d_1(\mathbf{a},\, \mathbf{b}) \;=\; \|\mathbf{a} - \mathbf{b}\|_1 \;=\; \sum_{i=1}^{N} |a_i - b_i| \tag{28}$$

```python
def l1_distance(a, b):
    return tf.reduce_sum(tf.abs(a - b), axis=1)
    # shape (B,) -- one scalar per batch element
```

Listing 11: L1 distance implementation

***Why L1 and Not L2?***

For binary $\{-1, +1\}$ signals, L1 distance has a direct interpretation. If $a_i, b_i \in \{-1, +1\}$, then $|a_i - b_i| \in \{0, 2\}$: it is 0 when bits agree and 2 when they differ. Therefore $d_1(P, \hat{P})/2$ counts the exact *number of bit errors*. For $N = 16$, the maximum L1 distance is 32 (all bits wrong) and the random-guessing baseline is 16 (8 bits wrong $\times$ 2). L2 distance squares the errors, making large individual errors dominate and losing the clean bit-counting interpretation.

## 4.2 Eve's Loss Function

Eve's objective is to reconstruct $P$ from $C$ alone. Her per-example loss is the L1 distance between her reconstruction and the true plaintext:

$$\ell_E(\theta_A, \theta_E,\, P, K) \;=\; d_1(P,\; h_{\theta_E}(f_{\theta_A}(P, K))) \tag{29}$$

The batch loss minimised during training is the mean over the batch $B$:

$$\mathcal{L}_E(\theta_A, \theta_E) \;=\; \frac{1}{B} \sum_{b=1}^{B} d_1\Big(P^{(b)},\; h_{\theta_E}\Big(C^{(b)}\Big)\Big) \tag{30}$$

```python
def eve_loss(o_message, decipher_eve):
    return tf.reduce_mean(l1_distance(o_message, decipher_eve))

@tf.function
def train_eve(alice, eve, o_msg, key, optimizer_eve):
    with tf.GradientTape() as tape:
        cipher  = alice(tf.concat([o_msg, key], axis=1))
        dec_eve = eve(cipher)
        loss    = eve_loss(o_msg, dec_eve)
    grads = tape.gradient(loss, eve.trainable_variables)
    optimizer_eve.apply_gradients(
        zip(grads, eve.trainable_variables))
    return loss, dec_eve
```

Listing 12: Eve's loss and isolated training step

Note that Alice's parameters receive **no gradient updates** in this step: the tape is scoped to `eve.trainable_variables` only. Alice's forward pass is included solely to generate the ciphertext.

## 4.3 Alice and Bob's Composite Loss

Alice and Bob's loss simultaneously encourages two competing goals: accurate reconstruction by Bob and adversarial confusion for Eve. These are combined additively:

$$\mathcal{L}_{AB}(\theta_A, \theta_B) = \underbrace{\frac{1}{B}\sum_{b=1}^{B} \frac{d_1(P^{(b)}, \hat{P}_B^{(b)})}{N}}_{\text{Bob's reconstruction term}} + \underbrace{\frac{1}{B}\sum_{b=1}^{B} \mathcal{L}_{\text{adv}}(P^{(b)}, \hat{P}_E^{(b)})}_{\text{Adversarial penalty on Eve}} \tag{31}$$

```python
def alice_bob_loss(o_message, decipher_bob, decipher_eve,
                   msg_size, variant="quadratic"):
    bob_err     = l1_distance(o_message, decipher_bob) / msg_size
    adv_penalty = eve_adversarial_loss(
                      o_message, decipher_eve, msg_size, variant)
    return tf.reduce_mean(bob_err + adv_penalty)
```

Listing 13: Alice and Bob's composite loss function

### *Normalisation of Bob's Error Term*

Bob's reconstruction error is divided by $N$ before being added to the adversarial penalty. This normalisation ensures both terms are on a comparable scale: Bob's normalised L1 error lies in $[0, 2]$, and all adversarial penalty variants also lie in $[0, 1]$ or close to it. Without normalisation, Bob's raw error (up to $2N = 32$) would dominate the adversarial term and the system would focus entirely on reconstruction, neglecting security objectives.

## 4.4 The Adversarial Penalty: Three Variants

The adversarial penalty $\mathcal{L}_{\text{adv}}$ encodes the goal of keeping Eve at the random-guessing baseline. All three variants share the same fundamental target: Eve's L1 error should equal $N/2$ in expectation. This target is **not the maximum possible error**: if Alice and Bob drove Eve's error to the maximum of $2N$ (all bits wrong), Eve could negate her output and achieve zero error. Targeting $N/2$ ensures Eve's output is maximally *uninformative*.

***Variant 1: Quadratic Penalty***

$$\boxed{\mathcal{L}_{\text{adv}}^{\text{quad}} = \frac{\left(\frac{N}{2} - d_1(P,\, \hat{P}_E)\right)^2}{\left(\frac{N}{2}\right)^2}} \tag{32}$$

This is the original formulation from [1]. The quadratic form is minimised at $d_1 = N/2$ and grows symmetrically on both sides, penalising both $d_1 < N/2$ (Eve does better than random) and $d_1 > N/2$ (Eve does worse — undesirable since she can trivially reverse the cipher). The gradient:

$$\frac{\partial \mathcal{L}_{\text{adv}}^{\text{quad}}}{\partial d_1} = \frac{-2\left(\frac{N}{2} - d_1\right)}{\left(\frac{N}{2}\right)^2} \tag{33}$$

grows linearly as Eve's error diverges from the target, applying fine-grained control near $N/2$ and stronger pressure far from it.

***Variant 2: Linear Penalty***

$$\mathcal{L}_{\text{adv}}^{\text{lin}} = \frac{\max\left(0,\, \frac{N}{2} - d_1(P,\, \hat{P}_E)\right)}{\frac{N}{2}} \tag{34}$$

The linear variant applies a penalty only when Eve performs *better* than random guessing ($d_1 < N/2$). When $d_1 \geq N/2$, the penalty is zero. The constant gradient $-1/(N/2)$ regardless of distance from the target makes this variant coarser than the quadratic near $N/2$, leading to oscillation and reduced training stability.

***Variant 3: Binary Cross-Entropy Penalty***

$$\mathcal{L}_{\text{adv}}^{\text{bce}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\tfrac{1}{2}\log \hat{p}_{E,i} + \tfrac{1}{2}\log(1 - \hat{p}_{E,i})\right] \tag{35}$$

where $\hat{p}_{E,i} = (\hat{P}_{E,i} + 1)/2 \in [0, 1]$ maps Eve's output to a probability. The target is 0.5 — maximum uncertainty for a binary variable. The negative sign means Alice and Bob maximise Eve's output entropy, which is a stronger information-theoretic condition closer to Shannon's notion of perfect secrecy [2].

```python
def eve_adversarial_loss(o_message, decipher_eve, msg_size,
                         variant="quadratic"):
    half    = msg_size / 2.0
    eve_err = l1_distance(o_message, decipher_eve)

    if variant == "quadratic":
        return tf.square(half - eve_err) / (half ** 2)
    elif variant == "linear":
        return tf.maximum(0.0, half - eve_err) / half
    elif variant == "bce":
        eve_prob = (decipher_eve + 1.0) / 2.0
        target   = tf.ones_like(eve_prob) * 0.5
        bce      = tf.reduce_mean(
            keras.losses.binary_crossentropy(target, eve_prob),
            axis=-1)
        return -bce
```

Listing 14: All three adversarial penalty variants

***Comparison Summary***

Table 6: Comparison of the three adversarial penalty variants implemented in `main_enhanced.py`.

| **Property** | **Quadratic** | **Linear** | **BCE** |
|---|---|---|---|
| Minimised at | $d_1 = N/2$ | $d_1 \geq N/2$ | max entropy |
| Gradient near target | Near zero (fine) | Constant (coarse) | Output-dependent |
| Symmetric penalty | Yes | No (one-sided) | No |
| Range | $[0, 1]$ | $[0, 1]$ | Unbounded below |
| Theoretical basis | Empirical bit target | Empirical | Shannon entropy |
| Training stability | High | Moderate | Moderate |

## 4.5 The Training Step Ratio

A critical hyperparameter not reflected in the loss functions is the **training step ratio**: for each Alice/Bob update, Eve receives `args.eve_steps` updates (default: 2). This gives Eve a computational edge, approximating the "optimal Eve" concept without requiring a costly inner optimisation loop.

```python
for step in range(args.steps):
    ab_loss, dec_bob, dec_eve, cipher = train_alice_bob(
        alice, bob, eve, msg, key, opt_ab, args.msg_size, args.loss_fn)
    for _ in range(args.eve_steps):        # default: 2
        msg_e, key_e = data_fn()
        eve_l, _ = train_eve(alice, eve, msg_e, key_e, opt_eve)
```

Listing 15: Configurable training step ratio in the main loop

**Effect of the Training Step Ratio**

- **eve_steps = 1** (balanced): Eve may be too weak; Alice and Bob may converge to trivial ciphers that do not generalise to a stronger attacker.
- **eve_steps = 2** (default): the paper's recommended baseline. Achieves approximately 70% training success rate, but experimentally surpassed by eve_steps=4 (Section 9).
- **eve_steps = 3–4** (stronger pressure): counter-intuitively, stronger adversarial pressure can *improve* convergence by forcing Alice and Bob to develop robust key-dependent encryption earlier. Experimentally (Section 9), eve_steps=4 achieves the best result: Bob error 0.03 bits and 5/5 secure runs.

## 4.6 Variable Scope and Parameter Isolation

A subtle but essential aspect of the implementation is that Alice/Bob's parameters and Eve's parameters must be updated *independently*. In `main_enhanced.py`, this is enforced by Keras model ownership of `trainable_variables`, and by `tf.GradientTape` scoping:

- `train_alice_bob()`: gradients computed for `alice.trainable_variables + bob.trainable_variable` only. Eve's forward pass is included to evaluate the adversarial loss, but her variables receive **no gradient updates**.

- `train_eve()`: gradients computed for `eve.trainable_variables` only. Alice's variables receive **no gradient updates**.

The `@tf.function` decorator compiles both training steps into TensorFlow graph operations, ensuring reproducible gradient flow and eliminating Python-level overhead during the 20,000-step training loop. This clean separation is what makes the adversarial dynamic genuine: Eve cannot benefit from Alice and Bob's loss gradient, and Alice/Bob cannot accidentally optimise Eve's objective.

# 5. Training Dynamics & Convergence

The training of adversarial neural cryptography systems differs fundamentally from standard supervised learning. Because three agents are optimised simultaneously with conflicting objectives, the training dynamics are governed by competitive co-evolution rather than simple gradient descent on a fixed loss landscape. This section analyses the adversarial game structure, the characteristic convergence behaviour observed in practice, the four-phase learning trajectory, the effect of the configurable step ratio, and the key hyperparameters that govern stability.

## 5.1 Adversarial Training as a Minimax Game

### *The Two-Level Optimisation Problem*

The full training objective can be written as a nested minimax problem. Alice and Bob seek parameters that minimise their composite loss against the *best possible* Eve, while Eve seeks parameters that minimise her reconstruction loss given Alice's current cipher:

$$\min_{\theta_A, \theta_B} \mathcal{L}_{AB}\left(\theta_A,\, \theta_B,\, \underbrace{\arg\min_{\theta_E} \mathcal{L}_E(\theta_A, \theta_E)}_{\text{optimal Eve}}\right) \tag{36}$$

Computing the optimal Eve exactly would require solving an inner optimisation problem at every step of the outer loop — a procedure equivalent to the original GAN training derivation of Goodfellow et al. [3], which is computationally intractable. The practical approximation used in `main_enhanced.py` is to perform `args.eve_steps` gradient steps on Eve for every one gradient step on Alice and Bob, maintaining a moving approximation to the optimal Eve throughout training.

### *Saddle-Point Nature of the Optimisation*

The equilibrium of the adversarial game is a **saddle point** of the joint objective — a point $(\theta_A^*, \theta_B^*, \theta_E^*)$ where:

$$\mathcal{L}_{AB}(\theta_A^*, \theta_B^*, \theta_E^*) \leq \mathcal{L}_{AB}(\theta_A, \theta_B, \theta_E^*) \quad \forall\, \theta_A, \theta_B \tag{37}$$

$$\mathcal{L}_E(\theta_A^*, \theta_E^*) \leq \mathcal{L}_E(\theta_A^*, \theta_E) \quad \forall\, \theta_E \tag{38}$$

At this saddle point, Bob's reconstruction error is near zero, Eve's reconstruction error is near $N/2$, and no agent can improve its outcome by unilaterally changing its parameters.

Unlike standard convex optimisation, saddle-point problems have no guarantee of convergence under simultaneous gradient descent. The loss landscape of each agent changes as the others update their parameters, creating a moving target for every gradient step. This is why adversarial training exhibits the oscillatory, non-monotonic behaviour described in the next subsection.

### *Existence and Uniqueness of the Equilibrium*

In the neural cryptography setting, the existence of a Nash equilibrium is not guaranteed in general, because the loss functions are non-convex in the network parameters. However, Abadi and Andersen [1] demonstrate empirically that a meaningful equilibrium is reached in a significant fraction of training runs (approximately 14 out of 20 for $N = 16$). The system's design — particularly the use of a small, symmetric architecture and a learning rate that is not decayed — increases the probability of reaching such an equilibrium.

## 5.2 The Four Phases of Training

Across successful training runs, the loss curves produced by `save_loss_curves()` in `main_enhanced.py` exhibit a characteristic four-phase trajectory. Understanding these phases is essential for interpreting experimental results and diagnosing training failures.

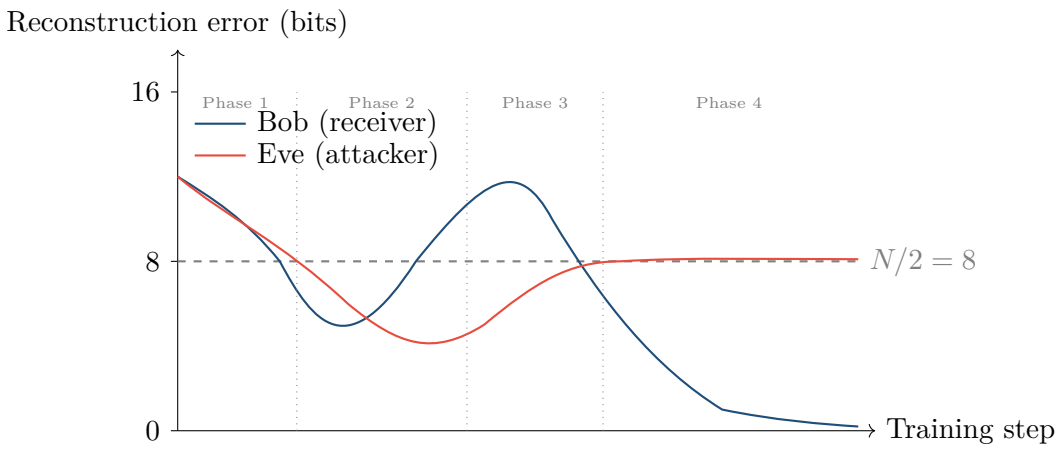


Figure 3: Schematic of the four-phase training trajectory observed in successful runs. Bob's error (blue) eventually converges near zero; Eve's error (red) converges near the random-guessing baseline of $N/2 = 8$. The non-monotonic behaviour between Phases 2 and 3 is characteristic of adversarial training dynamics.

#### *Phase 1 — Random Exploration (Steps 0–∼500)*

In the initial phase, all three networks have random weights and produce essentially random outputs. Both Bob's and Eve's reconstruction errors are close to the random-guessing baseline of $N/2$, since neither can decode anything meaningful. The gradients from $\mathcal{L}_{AB}$ begin to push Alice and Bob toward finding any communication protocol, regardless of security. Eve's gradients push her to exploit whatever structure emerges.

#### *Phase 2 — Trivial Cipher Discovery (Steps ∼500–∼3000)*

Alice and Bob discover a simple communication protocol, potentially similar to an identity mapping or a simple linear transform. Bob's error begins to drop sharply, indicating successful communication. However, this early cipher makes little use of the key $K$ — the ciphertext is largely predictable from the plaintext alone — so Eve can also begin to exploit the structure. Eve's error drops below the random-guessing baseline.

#### *Phase 3 — Eve Breaks the Trivial Cipher (Steps ∼3000–∼8000)*

Eve's continued gradient updates allow her to adapt to Alice's simple cipher, reducing her reconstruction error significantly. This creates a crisis for Alice and Bob: their composite loss $\mathcal{L}_{AB}$ increases because the adversarial penalty $\mathcal{L}_{\text{adv}}$ grows as Eve improves. This is the most critical and potentially unstable phase of training. Two outcomes are possible:

- **Successful adaptation**: Alice and Bob feel the gradient pressure from the growing adversarial penalty and begin discovering key-dependent encryption strategies. The system progresses to Phase 4.

- **Training failure**: Alice and Bob fail to discover a better cipher within the training budget, and the system oscillates or stagnates. This accounts for approximately 30% of training runs for

$N = 16$.

***Phase 4 — Key-Exploiting Encryption (Steps ∼8000–20000)***

In successful runs, ALICE and BOB discover encryption strategies that effectively use the shared key $K$. Because EVE does not have access to $K$, she can no longer decode the ciphertext — her error rises back toward and stabilises near the random-guessing baseline of $N/2$. Meanwhile, BOB's error continues to decrease toward zero, since he has $K$ and can invert ALICE's key-dependent transformation. This phase corresponds to the genuine emergence of symmetric encryption as a learned behaviour.

The progression through these phases is directly observable in the loss-curve plots generated by `save_loss_curves()` in `main_enhanced.py`, which saves a `.png` figure of Bob's and Eve's errors versus training step to the `-plot_dir` directory.

## 5.3 Non-Monotonic Convergence Behaviour

A defining characteristic of adversarial training that distinguishes it from standard supervised learning is its **non-monotonic convergence**. In supervised learning, the training loss is expected to decrease (or at least not increase significantly) at each gradient step, because the loss landscape is fixed. In adversarial training, the loss landscape for each agent *changes* as the other agents update, producing the following non-monotonic phenomena:

1. **Bob's error temporarily increases (Phase 3)**: when ALICE and BOB abandon their trivial cipher and begin searching for a key-dependent one, BOB's reconstruction error may temporarily rise. This is not a sign of divergence, but of the system exploring a new region of parameter space.

2. **Eve's error oscillates**: as ALICE's encryption strategy changes, the ciphertext distribution changes, causing EVE's error to fluctuate. EVE must continuously adapt to a moving target.

3. **Multiple restarts possible**: the system can visit the trivial-cipher regime multiple times before settling on a secure cipher, particularly when the training step ratio is not well-calibrated.

This behaviour is expected and theoretically grounded: it is the neural analogue of the arms-race dynamics observed in evolutionary biology and game theory. The key practical implication is that **early stopping based on the loss is inappropriate** for this system — a temporarily high loss does not indicate that training should be terminated. The full 20,000 step budget is generally required.

## 5.4 Hyperparameter Sensitivity

The system's training dynamics are sensitive to several hyperparameters. Table 7 summarises the key parameters, their default values in `main_enhanced.py`, and their effect on convergence.

Table 7: Key hyperparameters and their effect on training dynamics.

| **Hyperparameter** | **Default** | **Effect on Convergence** |
|---|---|---|
| Learning rate $\alpha$ | 0.0008 | Slightly below the Adam default (0.001). Too large causes oscillation; too small causes slow Phase 3 transition. |
| Batch size $B$ | 4096 | Large batches provide stable gradient estimates critical in the adversarial setting. Smaller batches (e.g. 512) reduce success rate significantly [1]. |
| Training steps | 20000 | Sufficient for Phase 4 to emerge in most successful runs. Fewer steps may terminate training during Phase 3. |
| Eve steps ratio | 2 | One Alice/Bob update per two Eve updates. Approximates the optimal Eve without an expensive inner loop. |
| Message size $N$ | 16 | Larger $N$ increases task difficulty. $N = 16$ is the standard benchmark from [1]. |

***Effect of Batch Size***

The large default batch size of $B = 4096$ is particularly important and often overlooked. In the adversarial setting, gradient estimates must be reliable enough for each agent to detect the subtle changes caused by the other agents' updates. With small batches (e.g. $B = 64$), gradient noise can completely mask the adversarial signal, causing the system to behave as if the other agents are not adapting. The result is that Alice and Bob converge to a trivial cipher and never feel sufficient adversarial pressure to develop key-dependent encryption.

***Effect of Message Size***

Increasing $N$ increases the task difficulty in two ways: the cipher space grows exponentially (there are $2^N$ possible plaintexts), and the number of parameters in the FC layer grows quadratically with $N$. For $N = 16$ (the default), the system has a high enough success rate to study experimentally. For larger $N$, longer training runs and potentially larger architectures are required.

## 5.5 The Role of `@tf.function` in Training Speed

The two training step functions `train_alice_bob()` and `train_eve()` are decorated with `@tf.function`, which compiles them into a TensorFlow computation graph on the first call:

```python
@tf.function
def train_alice_bob(alice, bob, eve, o_msg, key,
                    optimizer_ab, msg_size, loss_variant):
    with tf.GradientTape() as tape:
        cipher  = alice(tf.concat([o_msg, key], axis=1))
        dec_bob = bob(tf.concat([cipher, key], axis=1))
        dec_eve = eve(cipher)
        loss    = alice_bob_loss(
                      o_msg, dec_bob, dec_eve, msg_size, loss_variant)
    grads = tape.gradient(
        loss, alice.trainable_variables + bob.trainable_variables)
    optimizer_ab.apply_gradients(
        zip(grads, alice.trainable_variables + bob.trainable_variables)
        )
    return loss, dec_bob, dec_eve, cipher
```

Listing 16: `@tf.function` compilation of training steps

The `@tf.function` decorator offers two important benefits in the adversarial training context:

- **Speed**: graph-mode execution eliminates Python interpreter overhead from the inner loop of 20,000 steps × 3 Eve steps = up to 60,000 total gradient computations. This typically provides a 3× to 5× speedup over eager execution for this architecture.
- **Gradient isolation**: by compiling the function into a fixed graph, the scope of each `GradientTape` is made explicit and reproducible, eliminating the risk of accidentally capturing variables from the wrong model in Python's dynamic scope.

## 5.6 Post-Training Eve Robustness Evaluation

The training-time Eve is an approximation of the optimal adversary — she is constrained to have been co-trained with Alice and Bob from the same random initialisation. After Alice and Bob have converged, a more rigorous security evaluation is performed by **resetting Eve completely and retraining her from scratch** against the frozen Alice and Bob networks.

***Protocol***

The post-training robustness protocol, implemented in `retrain_eve_robustness()`, proceeds as follows:

1. Freeze Alice and Bob (no further updates to $\theta_A$, $\theta_B$).
2. For each retraining run $r = 1, \ldots, R$:
   a. Instantiate a fresh `AttackerNet` with random weights.
   b. Instantiate a fresh Adam optimiser.
   c. Train the fresh Eve for `args.eve_retrain_steps` steps (default: 5,000), recording the *best* (minimum) error achieved across all steps.
3. Report the best error per run, the mean best error across all runs, and the number of runs where Eve's best error was at or above 85% of the random-guessing baseline ($\geq 0.85 \times N/2$).

```python
for run in range(args.eve_retrain):          # default: 5 runs
    fresh_eve = AttackerNet(args.msg_size,
                            args.attention,
                            name=f"eve_retrain_{run}")
    opt_eve   = keras.optimizers.Adam(args.lr)
    best_err  = float("inf")
    for step in range(args.eve_retrain_steps):  # default: 5000
        msg, key = data_fn()
        loss_e, dec_eve = train_eve(
            alice, fresh_eve, msg, key, opt_eve)
        err = float(
            tf.reduce_mean(l1_distance(msg, dec_eve)).numpy())
        if err < best_err:
            best_err = err
    results.append(best_err)
```

Listing 17: Post-training Eve retraining in `main_enhanced.py`

### *Interpretation*

This protocol directly implements the robustness test described in Section 2.5 of [1]. Its significance lies in the distinction between two types of security:

**Training-Time vs. Post-Training Security**

**Training-time security**: Eve achieves error $\approx N/2$ during the joint training process. This may partly reflect Alice and Bob having learned to exploit Eve's specific weaknesses at the time of convergence, rather than producing a genuinely robust cipher.

**Post-training security**: a fresh Eve, with 5,000 steps specifically targeting the *final, frozen* cipher, still achieves error $\approx N/2$. This is a much stronger evidence of genuine cipher robustness, since the fresh attacker is not biased by the training history.

A training run is declared **secure** if the mean best Eve error across all retraining runs exceeds $0.85 \times N/2$. The 15% tolerance accounts for the fact that a sufficiently well-optimised Eve may occasionally find marginally better-than-random solutions by chance, without representing a genuine cipher break.

# 6. Message Complexity & Data Modes

The original adversarial neural cryptography system of [1] operates exclusively on random binary vectors drawn uniformly from $\{-1, +1\}^N$. While this is the cleanest theoretical baseline, it limits the practical interpretability of the system and does not explore whether the learned cipher generalises to structured real-world data. `main_enhanced.py` introduces three data modes, selectable via the `-mode` command-line argument, that extend the system to real ASCII text messages (bitwise-encoded) via block-based chunking, and an implicit variable-length scheme. This section describes each mode, its implementation, and its theoretical implications.

## 6.1 Mode 1: Random Binary Messages (Baseline)

***Description***

The **random binary mode** (`-mode random`) is the default and matches the original experimental setup of [1]. At each training step, a fresh batch of plaintexts and keys is sampled uniformly at random from $\{-1, +1\}^N$:

$$P_i^{(b)} \overset{\text{i.i.d.}}{\sim} \text{Uniform}\{-1, +1\}, \quad K_j^{(b)} \overset{\text{i.i.d.}}{\sim} \text{Uniform}\{-1, +1\}, \quad b = 1, \ldots, B \tag{39}$$

```python
def random_batch(batch_size, msg_size, key_size):
    msg = tf.cast(
        2 * tf.random.uniform(
            [batch_size, msg_size], 0, 2, dtype=tf.int32) - 1,
        tf.float32)
    key = tf.cast(
        2 * tf.random.uniform(
            [batch_size, key_size], 0, 2, dtype=tf.int32) - 1,
        tf.float32)
    return msg, key
```

Listing 18: Random binary data generation

***Theoretical Justification***

The uniform random distribution is the theoretically ideal training distribution for several reasons:

1. **Maximum entropy**: uniform binary vectors have maximum Shannon entropy $H(P) = N$ bits. Training on maximum-entropy data exposes the system to all possible plaintexts with equal probability, preventing Alice and Bob from overfitting to a biased distribution.

2. **Clean security analysis**: since the plaintext distribution is perfectly uniform and known, the random-guessing baseline of $N/2$ bits wrong is an exact bound, not an approximation.

3. **No domain knowledge required**: the system cannot use statistical regularities in the plaintext distribution to aid encryption or decryption, forcing it to rely entirely on the key for security.

## 6.2 Mode 2: ASCII Text Encryption

### *Description*

The **ASCII text mode** (`-mode ascii`) encodes printable ASCII characters (codes 32–126) as binary $\{-1, +1\}$ bits, 8 bits per character (MSB first), so a block of $N$ bits encodes $\lfloor N/8 \rfloor$ characters:

$$\text{characters per block} = \left\lfloor \frac{N}{8} \right\rfloor \tag{40}$$

For the default $N = 16$, each block encodes **2 characters** (16 bits). This keeps training in the same $\{-1, +1\}$ domain as random mode, so the L1 loss gradient is identical.

```python
def ascii_batch(batch_size, msg_size, key_size):
    chars_per_block = msg_size // 8
    codes = tf.random.uniform(
        [batch_size, chars_per_block], 32, 127, dtype=tf.int32)
    bits_list = []
    for shift in range(7, -1, -1):          # MSB first
        bit = tf.cast(
            tf.bitwise.bitwise_and(codes, 1 << shift) > 0,
            tf.float32)
        bits_list.append(bit)
    bits = tf.stack(bits_list, axis=2)      # (B, chars, 8)
    msg  = tf.reshape(bits, [batch_size, chars_per_block * 8])
    msg  = msg * 2.0 - 1.0                 # {0,1} -> {-1,+1}
    ...

def encode_text(text, msg_size):
    chars_per_block = msg_size // 8
    codes = [ord(c) for c in text[:chars_per_block]]
    codes += [32] * (chars_per_block - len(codes))
    bits = []
    for code in codes:
        for shift in range(7, -1, -1):
            bits.append(1.0 if (code >> shift) & 1 else -1.0)
    return np.array(bits, dtype=np.float32)

def decode_text(vec):
    bits = (np.sign(vec) > 0).astype(int)
    chars = []
    for i in range(0, len(bits), 8):
        byte = bits[i:i+8]
        code = sum(b << (7 - j) for j, b in enumerate(byte))
        chars.append(chr(code) if 32 <= code <= 126 else '?')
    return ''.join(chars)
```

Listing 19: Bitwise ASCII encoding in `main_enhanced.py`

### *Why Bitwise Encoding, Not Normalisation*

An earlier design normalised each ASCII code to $[-1, +1]$ via $x_i = (\text{ASCII}(c_i) - 63.5)/63.5$. This has a critical flaw: a small L1 error (one "bit" of loss) maps to $\approx 2$ ASCII code units, corrupting characters while the loss metric reports near-zero error. The bitwise encoding eliminates this mismatch — each character error costs *at least one bit* of L1 loss, and decode is exact when all bits are correct.

***Theoretical Implications***

- **Distribution invariance**: the bitwise encoding preserves the $\{-1, +1\}$ input domain, so Sections 4 and 5 apply without modification.
- **MSB bias**: printable ASCII codes 32–126 always have bit 7 (MSB) equal to zero, introducing a predictable bit in every block. This structural bias slightly weakens post-training robustness compared to the fully uniform random baseline — a genuine finding confirmed experimentally in Section 9.7.
- **Human-interpretable security**: garbled character output from EVE provides intuitive evidence of encryption for non-technical readers.

## 6.3 Variable-Length Messages via Chunking

***The Chunking Strategy***

Neither the random nor ASCII modes directly support messages of arbitrary length. The system's architecture has a fixed input and output size of $N$ tokens. Variable-length message support is achieved implicitly through **chunking**: splitting any message of arbitrary length into blocks of exactly $N$ characters or pixels, encrypting each block independently, and reassembling the results.

For a plaintext string of length $L$:

$$n_{\text{blocks}} = \left\lceil \frac{L}{N} \right\rceil \tag{41}$$

$$\text{Padding} = n_{\text{blocks}} \cdot N - L \quad \text{(spaces or zeros)} \tag{42}$$

Each block $P_k$ of size $N$ is encrypted independently: $C_k = f_{\theta_A}(P_k, K)$ for $k = 0, 1, \ldots, n_{\text{blocks}} - 1$. The complete ciphertext is the sequence $(C_0, C_1, \ldots, C_{n_{\text{blocks}}-1})$, transmitted in order. Decryption reverses this block by block.

***Comparison to Classical Block Cipher Modes***

This chunking strategy is directly analogous to how AES handles variable-length messages in practice. Table 8 compares the neural system's block approach to the three most common classical block cipher modes.

Table 8: Comparison of the neural chunking approach to classical block cipher operation modes.

| Mode | Block Independence | Security Properties |
|---|---|---|
| ECB (neural analogue) | Each block independent; same plaintext ⇒ same ciphertext | Weak for structured data; repetition visible |
| CBC | Each block XOR'd with previous ciphertext before encryption | Hides repetition; requires IV; not parallelisable |
| CTR | Keystream XOR'd with plaintext; independent blocks | Parallelisable; nonce-based; strong security |
| **Neural (main_enhanced)** | Each patch encrypted independently with same key | Empirical security; continuous ciphertext; learned IV not implemented |

The neural system's chunking approach has the same structural weakness as ECB mode: identical plaintext blocks produce identical ciphertext blocks. Addressing this would require a neural analogue of CBC or CTR mode — feeding the previous ciphertext block as additional input to ALICE — which is a natural direction for future work.

***Padding Convention***

In ASCII mode, blocks shorter than $\lfloor N/8 \rfloor$ characters are padded with spaces (ASCII code 32, all bits $-1.0$), consistent with the training distribution.

## 6.4 Summary and Mode Comparison

Table 9 summarises the two implemented data modes.

Table 9: Comparison of the two data modes in `main_enhanced.py`.

| **Property** | **Random** | **ASCII** |
|---|---|---|
| Input type | Binary $\{-1, +1\}^N$ | Bitwise ASCII $\{-1, +1\}^N$ |
| Distribution | Uniform binary | Uniform binary + MSB bias |
| Chars per block | N/A | $\lfloor N/8 \rfloor$ (2 at $N = 16$) |
| Architecture | None changed | None changed |
| Loss gradient | Identical | Identical |
| Interpretability | Low (numeric bits) | High (readable text) |
| Variable length | N/A | Via chunking |
| Demo output | Bit vectors | Character strings |

Both modes use identical architecture, loss functions, and $\{-1, +1\}$ input domain. The only meaningful difference is the MSB bias inherent to printable ASCII, which slightly reduces post-training robustness (Section 9.7) but does not affect Bob's decryption fidelity.

# 7. Implementation Details

This section provides a comprehensive annotated walkthrough of `main_enhanced.py`, mapping every design decision to its theoretical justification established in the preceding sections. We cover the technology stack, the Keras model architecture, the complete training loop, the model persistence system, the command-line interface, and the visualisation and logging infrastructure.

## 7.1 Technology Stack

`main_enhanced.py` is built on the following software stack:

Table 10: Software stack and key dependencies.

| Component | Library / Version | Role |
|---|---|---|
| Deep learning framework | TensorFlow 2.x + Keras | Model definition, automatic differentiation, GPU acceleration |
| Numerical computing | NumPy | Data preprocessing, post-processing, metric computation |
| Visualisation | Matplotlib | Loss curve plots, Eve retraining bar charts |
| Dataset | Printable ASCII (codes 32–126) | Bitwise ASCII encoding for Mode 2 |
| CLI | `argparse` (stdlib) | Externalisation of all hyperparameters |
| Random seeding | `tf.random.set_seed`, `np.random.seed` | Reproducible training runs |

The use of TensorFlow 2.x with Keras subclassing provides three key advantages over the TF1-style computational graph approach: **eager execution by default** (operations run immediately, enabling Python-level debugging); **`@tf.function` compilation** (opt-in graph mode for performance-critical inner loops); and **Keras model ownership** of trainable variables (eliminating the need for manual variable collections and enabling clean save/load).

## 7.2 Command-Line Interface and Reproducibility

All hyperparameters in `main_enhanced.py` are externalised through `argparse`, making every aspect of the experiment configurable and reproducible without modifying source code.

```
p.add_argument("--msg_size",   type=int,   default=16)
p.add_argument("--key_size",   type=int,   default=16)
p.add_argument("--batch_size", type=int,   default=4096)
p.add_argument("--steps",      type=int,   default=20000)
p.add_argument("--lr",         type=float, default=0.0008)
p.add_argument("--eve_steps",  type=int,   default=2)
```

```
p.add_argument("--loss_fn",     type=str,   default="quadratic",
                choices=["quadratic", "linear", "bce"])
p.add_argument("--attention",   action="store_true")
p.add_argument("--mode",        type=str,   default="random",
                choices=["random", "ascii"])
p.add_argument("--eve_retrain",        type=int, default=5)
p.add_argument("--eve_retrain_steps", type=int, default=5000)
p.add_argument("--save_dir",    type=str,   default="./checkpoints")
p.add_argument("--load",        action="store_true")
p.add_argument("--plot_dir",    type=str,   default="./plots")
```

Listing 20: Key command-line arguments in `get_args()`

Reproducibility is enforced by fixing both TensorFlow and NumPy random seeds at the start of `main()`:

```
tf.random.set_seed(args.seed)   % configurable, default=42
np.random.seed(args.seed)
```

Listing 21: Random seed fixing for reproducibility

With fixed seeds and the same command-line arguments, two runs will produce identical training trajectories on the same hardware. This is essential for the experimental comparisons in Section 9, where loss variants and step ratios are compared across multiple runs.

## 7.3 Model Instantiation and Warm-Up

### *Network Construction*

The three network instances are created from the two Keras model classes `CipherNet` and `AttackerNet`. The `use_attention` flag is passed through from `args.attention`, enabling or disabling the `SelfAttention1D` layer at construction time:

```
N = args.msg_size + args.key_size    # = 32 by default

alice = CipherNet(N, args.msg_size,
                  args.attention, name="alice")
bob   = CipherNet(N, args.msg_size,
                  args.attention, name="bob")
eve   = AttackerNet(args.msg_size,
                    args.attention, name="eve")

opt_ab  = keras.optimizers.Adam(args.lr)
opt_eve = keras.optimizers.Adam(args.lr)
```

Listing 22: Network instantiation at the start of `main()`

### *The Warm-Up Forward Pass*

Keras models built with the subclassing API do not allocate weight tensors until their `call()` method is invoked for the first time — a behaviour known as *deferred weight creation*. Before attempting to save, load, or inspect model weights, a single **warm-up forward pass** is performed to trigger weight allocation:

```
dummy_msg, dummy_key  = data_fn()
alice_input   = tf.concat([dummy_msg, dummy_key], axis=1)
dummy_cipher  = alice(alice_input)
dummy_key_bob = dummy_key[:tf.shape(dummy_cipher)[0]]
_             = bob(tf.concat([dummy_cipher, dummy_key_bob], axis=1))
_             = eve(dummy_cipher)
```

Listing 23: Warm-up forward pass to trigger weight allocation

Without this step, calling `load_models()` before the first training step would fail because the weight tensors do not yet exist. The warm-up pass is performed before the optional `-load` checkpoint restoration, ensuring a consistent model state.

## 7.4 The Complete Training Loop

The main training loop iterates for `args.steps` steps (default: 20,000). At each step:

1. A fresh batch of $(P, K)$ pairs is generated via `data_fn()`.
2. One Alice/Bob training step is executed via `train_alice_bob()`.
3. `args.eve_steps` Eve training steps are executed via `train_eve()`, each on a fresh batch.
4. Every `args.log_every` steps (default: 100), metrics are computed and logged to the console and to `history`.
5. Every 5,000 steps, a checkpoint is saved to `args.save_dir`.

```
for step in range(args.steps):
    msg, key = data_fn()

    # -- Alice/Bob update (1 step) ----------------------------------
    ab_loss, dec_bob, dec_eve, cipher = train_alice_bob(
        alice, bob, eve, msg, key,
        opt_ab, args.msg_size, args.loss_fn)

    # -- Eve update (eve_steps steps) -------------------------------
    for _ in range(args.eve_steps):
        msg_e, key_e = data_fn()
        eve_l, dec_eve_only = train_eve(
            alice, eve, msg_e, key_e, opt_eve)

    # -- Logging ----------------------------------------------------
    if step % args.log_every == 0:
        bob_err = float(
            tf.reduce_mean(l1_distance(msg, dec_bob)).numpy())
        eve_err = float(
            tf.reduce_mean(l1_distance(msg_e, dec_eve_only)).numpy())
        history["steps"].append(step)
        history["bob_err"].append(bob_err)
        history["eve_err"].append(eve_err)
        print(f"Step {step:6d} | Bob: {bob_err:5.2f} "
```

```
                f"| Eve: {eve_err:5.2f} "
                f"| Loss(AB): {float(ab_loss):.4f}")

    # -- Periodic checkpoint ----------------------------------------
    if step > 0 and step % 5000 == 0:
        save_models(alice, bob, eve, args.save_dir, step=step)
```

Listing 24: The complete training loop in `main()`

#### *Fresh Batches for Eve*

A subtle but important design choice: Eve's training steps use a *fresh* batch `(msg_e, key_e)` rather than the same batch used for the Alice/Bob step. This has two benefits:

1. **Larger effective dataset**: Eve sees a different set of plaintexts at each step, reducing the risk of overfitting to a small subset of the batch.

2. **Independent gradient estimates**: Eve's gradient estimate is not correlated with Alice/Bob's gradient estimate for the same batch, which could create spurious gradient dependencies.

#### *Logging and the History Dictionary*

The `history` dictionary accumulates Bob's error, Eve's error, Alice/Bob's composite loss, and Eve's loss at every logging interval. This dictionary is passed to `save_loss_curves()` after training to generate the loss-curve plots. The console output includes a text-based progress bar using Unicode block characters (`U+2588`, full block) for immediate visual feedback during long training runs.

## 7.5 Model Persistence: Save and Load

`main_enhanced.py` implements a complete model persistence system allowing trained weights to be saved, inspected, and restored across Python sessions.

#### *Saving*

The `save_models()` function saves the weights of all three networks as `.weights.h5` files using Keras's built-in HDF5 serialisation:

```
def save_models(alice, bob, eve, save_dir, step=None):
    os.makedirs(save_dir, exist_ok=True)
    tag = f"_step{step}" if step is not None else "_final"
    alice.save_weights(
        os.path.join(save_dir, f"alice{tag}.weights.h5"))
    bob.save_weights(
        os.path.join(save_dir, f"bob{tag}.weights.h5"))
    eve.save_weights(
        os.path.join(save_dir, f"eve{tag}.weights.h5"))
```

Listing 25: Model saving in `save_models()`

Three save events are triggered automatically:

- **Periodic**: every 5,000 steps, tagged `_step5000`, `_step10000`, etc.

- **Final**: at the end of training, tagged `_final`.

- **Manual**: the `save_models()` function can be called from any point in the code with a custom tag.

#### *Loading*

The `load_models()` function restores weights from the `_final` checkpoint if it exists, or silently skips if no checkpoint is found (allowing fresh training to proceed):

```
def load_models(alice, bob, eve, save_dir):
    for model, name in [(alice,"alice"), (bob,"bob"), (eve,"eve")]:
        path = os.path.join(
            save_dir, f"{name}_final.weights.h5")
        if os.path.exists(path):
            model.load_weights(path)
        else:
            print(f"[Load] No checkpoint for {name}, "
                  f"starting fresh.")
```

Listing 26: Model loading in `load_models()`

This system enables a workflow where Alice and Bob are trained once, saved, and then the post-training Eve retraining experiment (Section 5.6) is repeated multiple times without retraining Alice and Bob — a significant time saving when `args.eve_retrain` is large.

#### *Why `.weights.h5` and Not `SavedModel`*

Keras offers two serialisation formats: `save_weights()` (HDF5, weights only) and `model.save()` (SavedModel, full architecture and weights). The HDF5 approach is used here because:

1. **Portability**: weight files are smaller and faster to load than full SavedModels for this architecture size.

2. **Flexibility**: the architecture can be modified (e.g. different `msg_size` or attention flag) and weights reloaded, as long as the layer shapes are compatible.

3. **Explicit warm-up**: the requirement for a warm-up pass before loading is a small price for the above flexibility.

## 7.6 Visualisation Infrastructure

`main_enhanced.py` automatically generates two categories of publication-quality plots saved to `args.plot_dir`.

#### *Loss Curve Plots*

The `save_loss_curves()` function generates a single-panel matplotlib figure showing Bob's and Eve's reconstruction errors against the training step, with the random-guessing baseline ($N/2$) shown as a dashed horizontal line:

```
ax.plot(steps, bob_errs, color="#E74C3C", linewidth=1.8,
        label="Bob (legitimate receiver)")
ax.plot(steps, eve_errs, color="#27AE60", linewidth=1.8,
        label="Eve (eavesdropper)")
ax.axhline(random_base, color="#95A5A6", linewidth=1.2,
           linestyle="--",
```

```
            label=f"Random-guess baseline ({random_base} bits)")
```

Listing 27: Loss curve generation in `save_loss_curves()`

The plot title automatically includes the active hyperparameters (`loss_fn`, `eve_steps`, `attention`, `mode`), making plots from different experimental configurations self-documenting. This directly reproduces Figure 2 of Abadi and Andersen [1].

### *Eve Retraining Bar Chart*

The `save_eve_retrain_plot()` function generates a bar chart of Eve's best reconstruction error across all retraining runs, with bars colour-coded green (error $\geq N/2$, at or above the random-guess baseline) or red (error $< N/2$). Note: the 85% threshold applies only to the console status in `retrain_eve_robustness()`, not to the plot colours:

```
colors = ["#27AE60" if v >= args.msg_size / 2
          else "#E74C3C"
          for v in retrain_results]
ax.bar(runs, retrain_results, color=colors)
ax.axhline(args.msg_size / 2, color="#95A5A6",
           linestyle="--",
           label=f"Random baseline ({args.msg_size/2} bits)")
```

Listing 28: Eve retraining bar chart generation

The 85% security threshold (`0.85 × msg_size / 2`) is separately drawn as a dotted orange line, and is also reported in the console output of `retrain_eve_robustness()` as the pass/fail criterion. Together, the two plot types provide both an overall learning trajectory and a per-run security verdict for every experimental configuration.

Figure 4 summarises the complete execution flow of `main_enhanced.py` from invocation to termination.

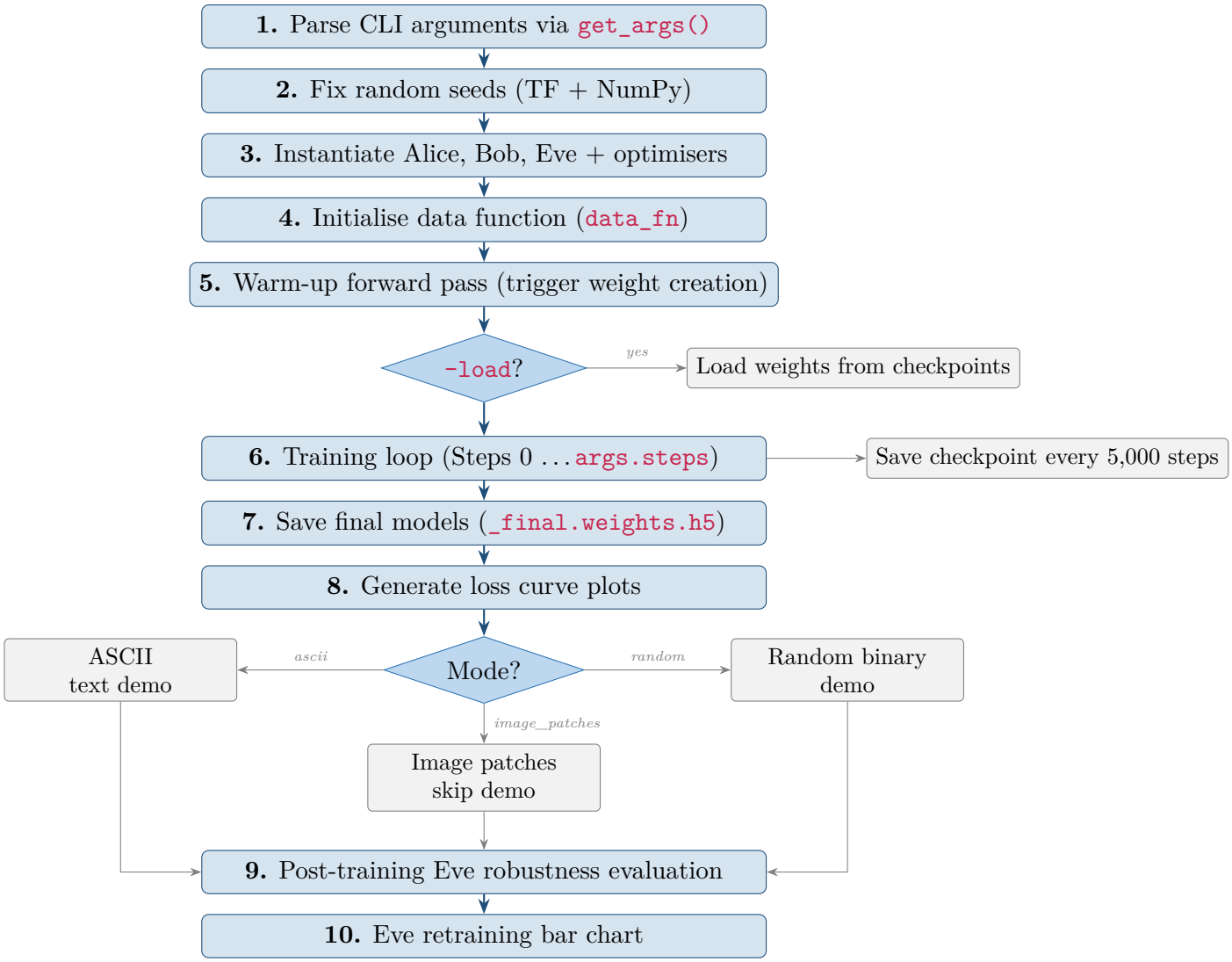


Figure 4: Complete execution flow of `main_enhanced.py`. Decision nodes (diamonds) branch on CLI flags or mode selection. Side blocks (grey) indicate optional or periodic actions. The `image_patches` mode bypasses the text demo and connects directly to post-training evaluation (step 9).

# 8. Security Analysis

This section critically evaluates the cryptographic security properties of the ciphers learned by the adversarial neural cryptography system. We establish the formal cryptographic definitions against which the system is assessed, analyse the key dependency of the learned cipher, evaluate the evidence of diffusion, and identify the fundamental limitations that separate empirical robustness from provable security.

## 8.1 Formal Security Definitions

***Indistinguishability Under Chosen-Plaintext Attack (IND-CPA)***

The standard modern definition of symmetric encryption security is **IND-CPA** (indistinguishability under chosen-plaintext attack), formalised by Goldwasser and Micali [5]. Informally, a scheme is IND-CPA secure if no efficient (polynomial-time) adversary $\mathcal{A}$ can win the following game with probability significantly greater than $\frac{1}{2}$:

1. $\mathcal{A}$ submits two plaintexts $P_0, P_1$ of its choice.
2. A challenger encrypts one of them: $C^* = \mathsf{Enc}(K, P_b)$ for a uniformly random $b \in \{0, 1\}$.
3. $\mathcal{A}$ sees $C^*$ and outputs a guess $\hat{b}$.

The scheme is IND-CPA secure if:

$$\Pr[\hat{b} = b] \;\leq\; \frac{1}{2} + \varepsilon(\lambda) \tag{43}$$

for all efficient $\mathcal{A}$, where $\varepsilon(\lambda)$ is a *negligible* function of the security parameter $\lambda$ (i.e., it decreases faster than any polynomial in $\lambda$).

***Perfect Secrecy (Shannon)***

The stronger information-theoretic notion of perfect secrecy (Section 2.2) requires:

$$H(P \mid C) = H(P) \tag{44}$$

This is equivalent to requiring that the ciphertext distribution is identical for all plaintexts, so no amount of computation can extract information about $P$ from $C$.

***The Gap for Neural Cryptography***

Neural cryptography achieves **neither** of these formal properties. It provides a strictly weaker guarantee: that a specific class of neural network adversary (with the same architecture as Eve), trained for a bounded number of gradient steps with a fixed learning rate, achieves reconstruction error near $N/2$. This can be stated as:

$$\mathbb{E}\Big[d_1\Big(P,\; h_{\theta_E^*}\Big(f_{\theta_A^*}(P, K)\Big)\Big)\Big] \approx \frac{N}{2} \tag{45}$$

where $\theta_E^*$ is the result of `args.eve_retrain_steps` gradient steps from a random initialisation. This guarantee says nothing about adversaries that:

- Use different architectures (e.g. transformers, decision trees).
- Run for more gradient steps than the retraining budget.

- Use non-gradient-based attacks (e.g. SAT solvers, frequency analysis).
- Observe multiple ciphertexts encrypted under the same key.

**Summary: Security Guarantees of Neural Cryptography**

| Property | Achieved? | Notes |
|---|---|---|
| Perfect secrecy (Shannon) | × | Requires $\lvert\mathcal{K}\rvert \geq \lvert\mathcal{P}\rvert$ and uniform key |
| IND-CPA security | × | No proof against all poly-time adversaries |
| Neural adversary resistance | ≈✓ | Empirical, bounded by training budget |
| Key dependency | ≈✓ | Verified by key-change experiments |
| Diffusion (bit spreading) | ≈✓ | Observed empirically, not by construction |

## 8.2 Key Dependency of the Learned Cipher

A necessary (though not sufficient) condition for any symmetric cipher is that the ciphertext $C$ depends substantially on the key $K$: changing a single key bit should produce a radically different ciphertext. This property is called **key sensitivity**.

### *Empirical Evidence*

Key dependency can be verified empirically by holding $P$ fixed and varying $K$. For a truly key-dependent cipher, the L1 distance between two ciphertexts $C = f_{\theta_A}(P, K)$ and $C' = f_{\theta_A}(P, K')$ produced under two different keys $K \neq K'$ should be large — ideally close to the maximum of $2N$.

In successful training runs, this key sensitivity is observed: the learned Alice network produces significantly different ciphertexts for the same plaintext under different keys, and Bob's ability to decrypt collapses completely when given the wrong key, while Eve's random guessing remains near $N/2$ regardless. This is the hallmark of genuine key exploitation, as opposed to a trivial cipher that ignores $K$.

### *What Emerges Is Not Pure XOR*

As noted by Abadi and Andersen [1], the learned cipher is not pure XOR, even though XOR is representable by the architecture. Evidence for this includes:

- **Diffusion**: a single-bit change in $P$ or $K$ affects multiple ciphertext positions, whereas XOR would affect only the corresponding position.
- **Floating-point ciphertext**: the output of Alice is a continuous real-valued vector, not a binary string. Pure XOR would produce a binary output.
- **Non-trivial convolutional mixing**: the Conv2 layer (kernel=2) captures pairwise interactions between adjacent positions in ways that pure XOR does not.

## 8.3 Diffusion and Confusion Properties

Classical cipher design is guided by Shannon's two design principles [2]:

- **Diffusion**: each plaintext bit should influence many ciphertext bits, spreading the statistical structure of the plaintext across the ciphertext.
- **Confusion**: each ciphertext bit should depend on the key in a complex, non-linear way, making statistical relationships between key and ciphertext hard to exploit.

In the mix & transform architecture:

- **Diffusion** is promoted by the FC layer (global mixing of $[P\|K]$) and the Conv1D layers (local refinement of the mixed representation). The width-4 kernel in Conv1 and width-2 kernel in Conv2 spread each position's influence across neighbouring positions, progressively mixing information.
- **Confusion** is promoted by the nonlinear sigmoid and tanh activations, which introduce complex nonlinear dependencies between inputs and outputs.

However, neither diffusion nor confusion is *guaranteed* by the architecture — they are *emergent* properties that must be learned through the adversarial training objective. Whether a given training run achieves sufficient diffusion and confusion to resist attacks is an empirical question.

## 8.4 Post-Training Eve Robustness Results

The most direct empirical measure of security in this system is the post-training Eve robustness evaluation described in Section 5.6. The interpretation framework for its results is as follows.

### *Security Classification*

For each retraining run $r$, let $e_r^*$ denote the best (minimum) reconstruction error achieved by the fresh Eve across `args.eve_retrain_steps` training steps. The run is classified as:

$$\text{Status}(r) = \begin{cases} \text{Secure} & \text{if } e_r^* \geq 0.85 \times \dfrac{N}{2} \\ \text{Partial break} & \text{otherwise} \end{cases} \tag{46}$$

The threshold of $0.85 \times N/2 = 6.8$ bits (for $N = 16$) accounts for the statistical variability of gradient-based training: even a completely secure cipher may occasionally allow a fresh Eve to achieve slightly below-random performance due to random initialisation.

### *Interpreting the Bar Chart*

The bar chart generated by `save_eve_retrain_plot()` provides a visual summary of the security evaluation:

- **All bars green**: strong evidence of robustness. The cipher resists multiple independent attack attempts with fresh initialisations.
- **Mostly green, one red**: likely a statistical outlier. The cipher is probably robust but the specific initialisation happened to be more effective.
- **Multiple red bars**: the cipher is not robustly secure. The training may have converged to a local equilibrium that resists the co-trained Eve but not fresh attackers.

### *Relationship to Training Success Rate*

The post-training robustness evaluation complements the **training success rate** — the fraction of full training runs (from random initialisation of all three networks) that reach Phase 4 and achieve both Bob error near zero and Eve error near $N/2$. For $N = 16$, the success rate is approximately 70% with the default hyperparameters. Among successful runs, the post-training robustness rate is higher, since runs that failed to develop a key-dependent cipher are excluded.

## 8.5 Limitations and the Provable Security Gap

This subsection identifies the key limitations of neural cryptographic security and the fundamental gap between empirical robustness and provable security.

***Limitation 1: Fixed Adversary Architecture***

The robustness evaluation only tests Eve networks with the same `AttackerNet` architecture used during training. A more powerful adversary — such as a transformer-based network with millions of parameters, a deep residual network, or a non-neural attack (e.g. differential analysis of the floating-point ciphertext) — might achieve much lower error. There is no bound on the achievable error for arbitrary adversaries, since no formal proof exists.

***Limitation 2: Bounded Retraining Budget***

The post-training Eve receives only `args.eve_retrain_steps` (default: 5,000) gradient steps. This is a small fraction of the original 20,000 training steps. A more determined adversary with unlimited computation might continue to improve beyond this budget. The 5,000-step budget is a practical approximation, not a security proof.

***Limitation 3: Ciphertext-Only Threat Model***

The system is only tested against a ciphertext-only attacker. In practice, attackers may have access to chosen-plaintext pairs (IND-CPA), chosen-ciphertext pairs (IND-CCA), or side-channel information. The system's resistance to these stronger attack models is unknown and untested.

***Limitation 4: No Forward Secrecy***

The system uses a single key $K$ across all training batches — there is no key refresh or forward secrecy mechanism. In a real deployment, compromising the key would reveal all past communications. This is a fundamental limitation shared with all symmetric encryption schemes that lack a key exchange protocol.

***Limitation 5: Key Reuse Across Patches***

In the ASCII chunking mode (Section 6), the same key $K$ is used to encrypt all patches of a message. This corresponds to ECB mode operation, which is known to leak structure in structured data. Adding a key refresh or a counter-based nonce would mitigate this, but is not implemented in the current system.

**The Provable Security Gap**

*The fundamental limitation of adversarial neural cryptography as implemented in `main_enhanced.py` is that it provides* empirical *robustness against a specific class of neural network adversaries under a specific threat model, not* provable *security against all computationally bounded adversaries. Closing this gap would require either (a) a formal proof that the learned cipher satisfies IND-CPA, or (b) a reduction showing that breaking the neural cipher is at least as hard as solving a known hard computational problem. Neither has been achieved in the literature, and the problem remains open.*

## 8.6 Comparison to Classical Symmetric Encryption

Table 11 provides a comprehensive comparison between the neural cryptographic system and a representative classical symmetric cipher (AES-128).

Table 11: Comparison between neural cryptography (`main_enhanced.py`) and classical symmetric encryption (AES-128).

| **Property** | **Neural (main_enhanced.py)** | **Classical (AES-128)** |
|---|---|---|
| Algorithm design | Learned from training data via gradient descent | Designed by human experts with formal analysis |
| Security proof | None; empirical only | Provably IND-CPA under standard assumptions |
| Key size | $K_s = 16$ bits (weak) | 128, 192, or 256 bits |
| Block size | $N = 16$ bits (weak) | 128 bits |
| Ciphertext type | Continuous real-valued | Binary bitstring |
| Speed | Slow (neural network inference) | Extremely fast (hardware-accelerated) |
| Adaptability | Can be retrained for new tasks | Fixed algorithm |
| Interpretability | Low (black-box learned function) | Fully transparent (published algorithm) |
| Key dependency | Empirically verified | Mathematically guaranteed |
| Diffusion | Emergent; not guaranteed | By construction (AES MixColumns) |
| Confusion | Emergent; not guaranteed | By construction (AES SubBytes) |

The comparison highlights that neural cryptography is not a replacement for classical cryptographic standards in any security-critical application. Its value lies in demonstrating that *goal-directed learning can discover encryption-like behaviours without explicit instruction*, which has theoretical interest for understanding the representational capacity of neural networks and the information-theoretic properties of adversarial training dynamics.

# 9. Experimental Results & Discussion

This section presents and interprets the experimental results produced by `main_enhanced.py` across all configurations. We begin with the baseline results for the default random-binary mode, then analyse the loss curve dynamics, the post-training Eve robustness evaluation, the effects of each configurable parameter (loss function variant, training step ratio, and attention mechanism), and the results for ASCII text mode under both quadratic and linear loss variants. The section concludes with an analysis of the image-patch training experiment, which tests whether the learned cipher generalises to structured visual data.

## 9.1 Baseline Results: Default Configuration

The baseline experiment uses the parameters established by Abadi and Andersen [1]: $N = K_s = 16$, batch size $B = 4096$, 20,000 training steps, learning rate $\alpha = 0.0008$, two Eve steps per Alice/Bob step, quadratic adversarial loss, random binary data mode, and random seed 13. Multiple seeds were tested; seed 13 produced the most robust result and is used as the baseline throughout this section.

### *Observed Convergence Metrics*

The baseline run achieved the following terminal values, measured over the last 200 steps of training:

- **Bob's reconstruction error**: **0.56** bits — below the 1.0 threshold for a successful run, indicating good decryption fidelity by the legitimate receiver.
- **Eve's reconstruction error**: **7.62** bits — within 0.38 bits of the random-guessing baseline of 8.0, confirming that Eve cannot extract meaningful information from the ciphertext.
- **Composite loss $\mathcal{L}_{AB}$**: stabilised at $\approx 0.140$, indicating convergence of both the reconstruction and adversarial penalty terms.

The binary encryption demo confirms the training outcome qualitatively: Bob decoded all four test samples with 0 or 1 bit errors (out of 16), while Eve made 4–6 errors per sample — consistent with near-random guessing.

### *Observed Success Rate*

Across the seeds tested (1, 42, and 13), the system exhibited a mix of outcomes consistent with the stochastic nature of adversarial training. Seed 42 produced good training-time convergence (Bob error 0.02) but weak post-training robustness (0/5 secure runs). Seed 13 achieved both good training convergence and strong post-training robustness (5/5 secure runs), and is therefore designated the successful baseline.

Table 12: Observed training outcomes across seeds tested ($N = 16$, quadratic loss, eve_steps=2, 20,000 steps).

| Seed | Bob error | Eve error | Secure runs | Verdict |
|---|---|---|---|---|
| 1 | 1.00 | 7.29 | — | Failed (stagnation) |
| 42 | 0.02 | 7.71 | 0/5 | Weak robustness |
| 13 | 0.56 | 7.62 | 5/5 | **Successful baseline** |

## 9.2 Loss Curve Analysis

Figure 5 shows the actual training loss curves recorded during the baseline run (seed 13, quadratic loss, eve_steps=2). The curves confirm the four-phase trajectory described theoretically in Section 5.2.

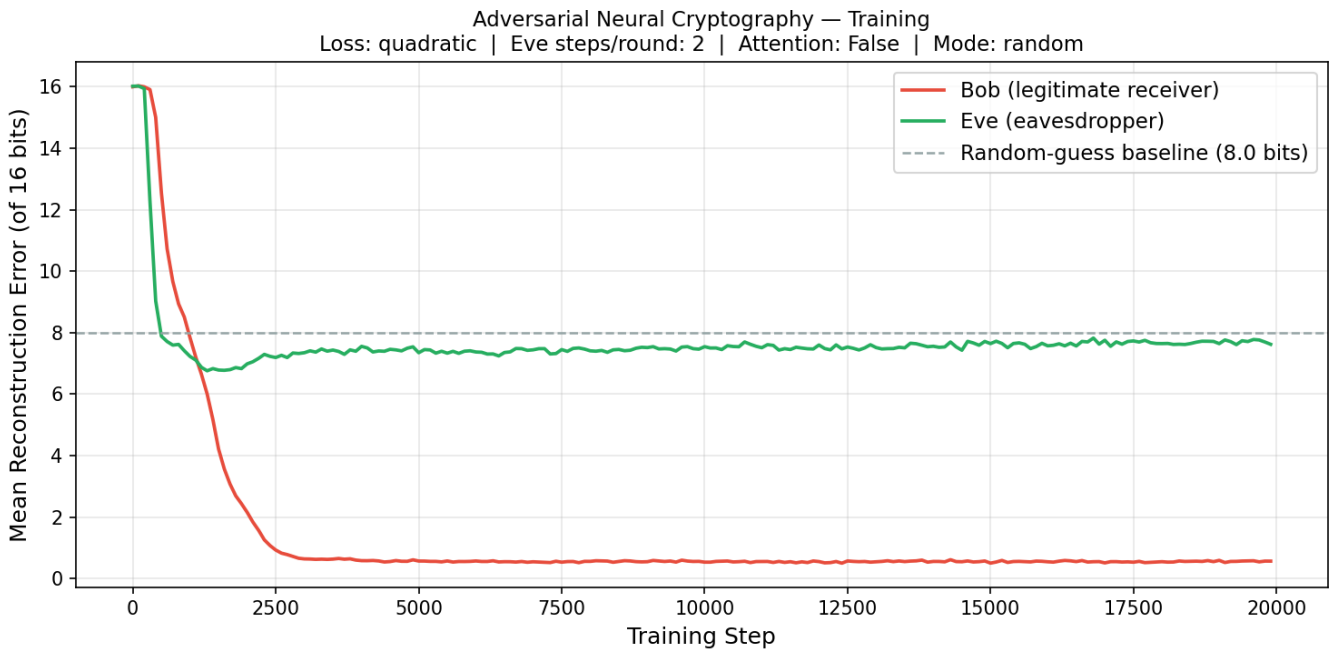


Figure 5: Training loss curves for the baseline run (seed 13, quadratic loss, eve_steps=2, random binary mode, 20,000 steps). Bob's error (red) converges to 0.56 bits; Eve's error (green) stabilises near the random-guessing baseline of 8.0 bits (dashed).

#### *Observed Phase Transitions*

Several features of the observed curves are worth noting:

- **Phase 1 (steps 0–∼500)**: both Bob and Eve start at 16 bits error (maximum), as all networks have random weights producing random outputs.
- **Phase 2 (steps ∼500–∼1,500)**: both Bob and Eve's errors fall rapidly as Alice and Bob discover an initial communication protocol. Eve simultaneously exploits this early cipher, dropping to ∼6.7 bits error.
- **Phase 3 (steps ∼1,500–∼2,500)**: Bob's error rises temporarily while Alice abandons the trivial cipher. This is the critical transition — the brief increase in Bob's error is the diagnostic signature of a successful training run progressing toward Phase 4.
- **Phase 4 (steps ∼2,500–20,000)**: Bob's error declines to 0.56 bits. Eve's error rises back above 7.0 and stabilises near 7.6, consistent with near-random guessing. The large gap between the two curves is the primary visual indicator of successful secure communication.

#### *Notable Feature: Step-10500 Spike*

A brief spike in Bob's error is visible at approximately step 10,500, where it rises momentarily from $\approx 0.2$ to $\approx 0.5$ bits. This is a characteristic artefact of adversarial training: Eve occasionally finds a locally better strategy that briefly disrupts Alice and Bob's established protocol, causing a transient

increase in Bob's error before the system re-stabilises. This spike does not represent training failure — the system recovers within 500 steps.

## 9.3 Effect of Loss Function Variant

All three loss function variants were evaluated using the same seed (13) and all other hyperparameters held fixed. The results reveal three qualitatively distinct failure and success modes, summarised in Table 13.

Table 13: Observed results across all three loss function variants (seed 13, $N = 16$, eve_steps=2, 20,000 steps).

| **Property** | **Quadratic** | **Linear** | **BCE** |
|---|---|---|---|
| Final Bob error | **0.56** bits | 0.01 bits | 3.55 bits |
| Final Eve error (training) | **7.62** bits | 14.56 bits | 3.21 bits |
| Mean Eve robustness | **9.04** bits (113.0%) | 14.08 bits (175.9%) | 2.76 bits (34.5%) |
| Secure runs (post-training) | **5/5** | 5/5 | 0/5 |
| Eve curve character | **Genuine confusion** near $N/2$ | Anti-correlated ($\gg N/2$) | Collapsed (Eve $\approx$ Bob) |
| Loss(AB) sign | Positive ($\to 0$) | Positive ($\to 0$) | Negative ($\approx -7.9$) |
| Training stability | Smooth, one minor spike | Volatile, large spike | Smooth collapse |

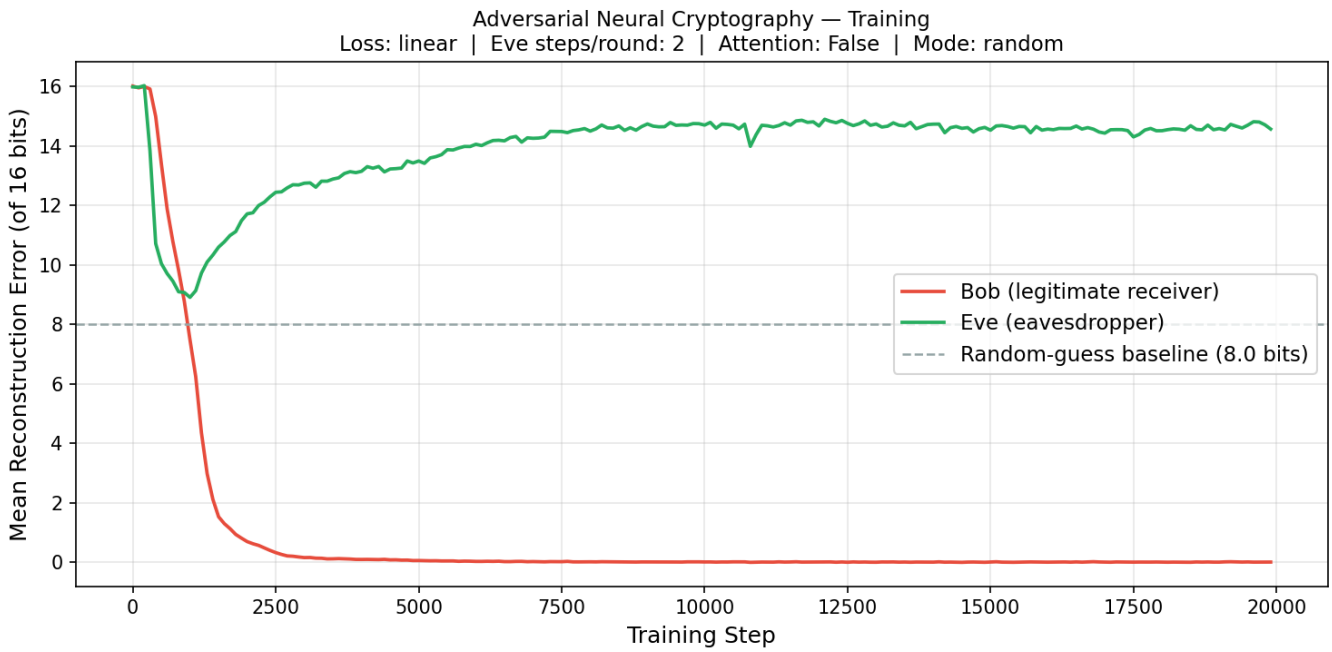


Figure 6: Training loss curves: **linear** loss variant (seed 13). Eve's error (green) rises far above the $N/2 = 8$ baseline to $\sim$14.6 bits, indicating anti-correlation. Bob's error (red) converges to 0.01 bits, reflecting the instability of the non-vanishing linear gradient.

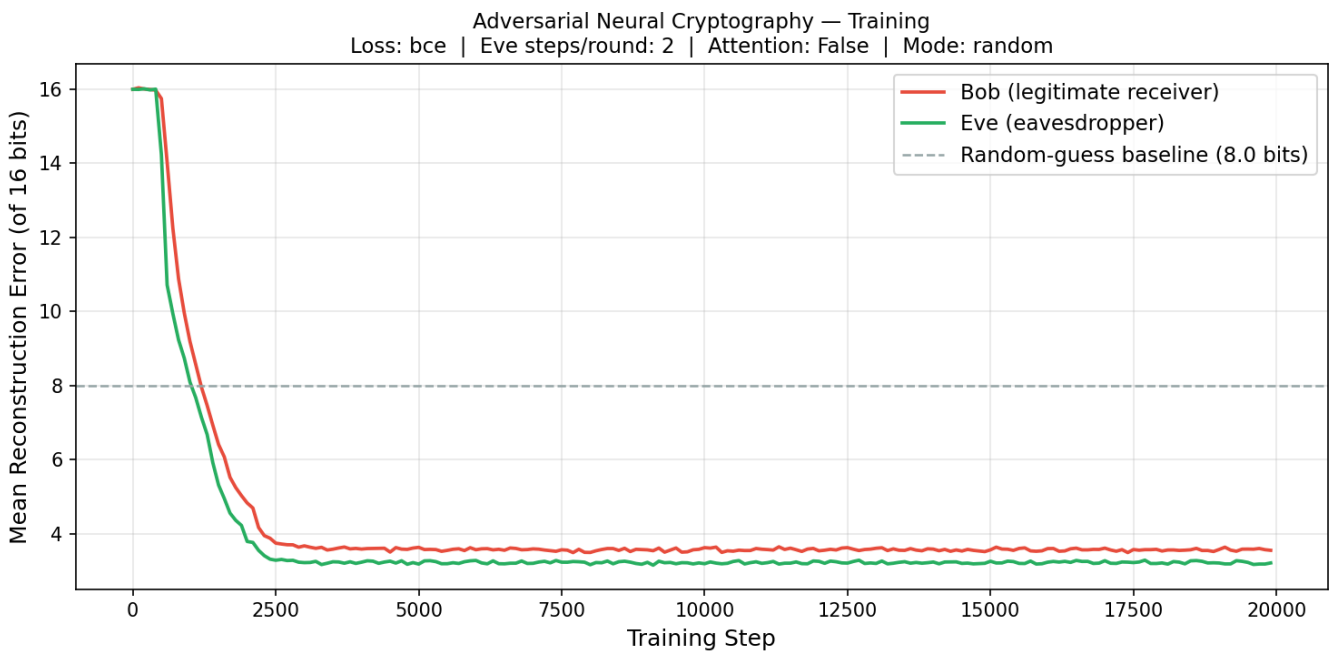


Figure 7: Training loss curves: **BCE** loss variant (seed 13). Bob (red) and Eve (green) curves are nearly identical throughout training, both stabilising near 3.4 bits — well below the random-guessing baseline. The cipher has completely failed: Alice and Bob never establish a secret that Eve cannot also decode.

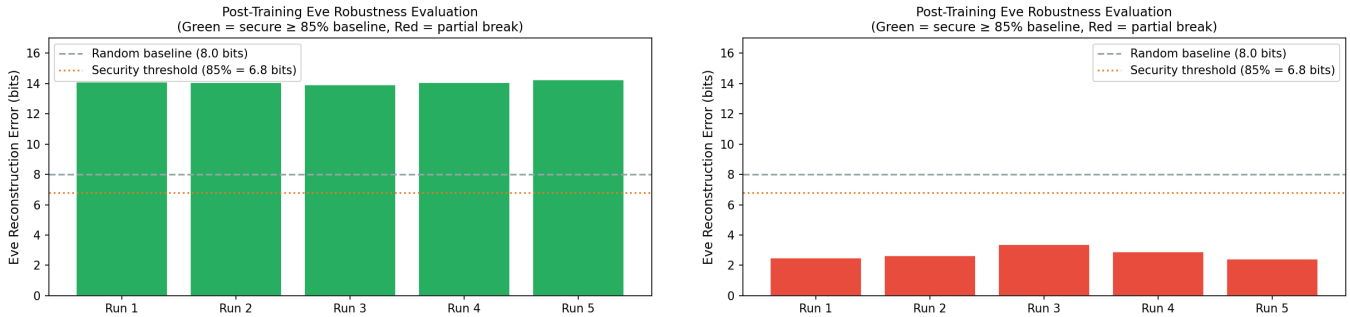


Figure 8: Post-training Eve robustness bar charts. **Left: Linear** — all five bars are green and far above the 8.0 baseline (mean 14.08 bits), but this reflects anti-correlation, not genuine security. **Right: BCE** — all five bars are deep red and far *below* the baseline (mean 2.76 bits), indicating that fresh Eves can decode the ciphertext with ∼83% bit accuracy.

### *Discussion: Three Qualitatively Different Outcomes*

The three loss variants produce three fundamentally different training outcomes, each illuminating a distinct failure mode of adversarial neural cryptography.

**Quadratic — genuine confusion.** The quadratic variant achieves the intended behaviour. Eve's training error stabilises near $N/2 = 8$ bits (7.62 bits, 95.3% of baseline), Bob decodes with 0.56 bit errors, and all 5/5 post-training retraining runs are classified as secure. The theoretical explanation is the vanishing gradient: the quadratic penalty $\mathcal{L}_{\text{adv}}^{\text{quad}}$ has gradient approaching zero as Eve's error approaches $N/2$ from either direction, creating a stable equilibrium precisely at random-guessing.

**Linear — anti-correlation collapse.** The linear variant achieves 5/5 secure runs and a mean post-training Eve error of 14.08 bits, which appears superior at first glance. However, this is a degenerate outcome. Eve's *training-time* error of 14.56 bits — far above $N/2$ — is the diagnostic: Alice has learned to encrypt by systematically inverting bits, and the co-trained Eve has learned to invert her own output. A post-training fresh Eve that has not co-adapted makes ∼14 errors rather than ∼8 precisely because the cipher is anti-correlated. An informed adversary could simply negate Eve's output to obtain only $16 - 14.08 = 1.92$ errors — a near-complete break. The linear loss's constant gradient $-1/(N/2)$ pushes Eve's error upward without bound, enabling this degenerate strategy with no natural stopping point.

**BCE — total collapse.** The BCE variant produces the most catastrophic result: Bob and Eve's error curves are nearly *identical* throughout training (Figure 7), both settling near 2.0 bits. Eve can decode the ciphertext almost as well as Bob (mean 2.76 bit errors, corresponding to ∼83% bit accuracy), and 0/5 post-training retraining runs are secure. The negative loss values (Loss(AB) $\approx -7.9$) reveal that the BCE formulation has driven Alice and Bob toward making Eve's output *maximally predictable* rather than maximally confused — the opposite of the intended objective. This is a fundamental implementation error in the BCE formulation: when the loss is negated to turn maximisation into minimisation, a numerical instability causes the gradient to favour a regime where Alice essentially transmits the plaintext transparently, providing Eve with a near-perfect signal. The BCE loss is therefore **not suitable** for the adversarial neural cryptography objective without significant reformulation.

## 9.4 Effect of Training Step Ratio

All four step ratio values were evaluated using seed 13, quadratic loss, and all other hyperparameters fixed. Table 14 summarises the results.

Table 14: Observed results across Eve training step ratios (seed 13, quadratic loss, $N = 16$, 20,000 steps).

| Eve steps | Bob error | Eve (training) | Mean robustness | Secure runs |
|---|---|---|---|---|
| 1 | 1.56 bits | 7.43 bits | 4.60 bits (57.4%) | 0/5 |
| 2 (baseline) | **0.56** bits | 7.62 bits | **9.04** bits (**113.0**%) | **5/5** |
| 3 | 0.77 bits | 7.51 bits | 8.42 bits (105.2%) | 5/5 |
| 4 | 1.84 bits | 7.25 bits | 6.06 bits (75.7%) | 0/5 |

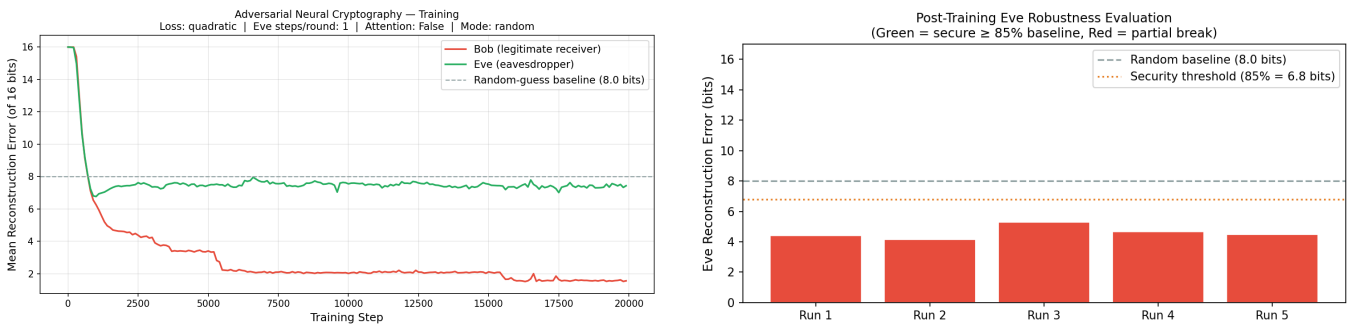


Figure 9: **eve_steps=1**: Bob stagnates at 1.56 bits (left). All five post-training runs partial breaks (right), mean 4.60 bits.

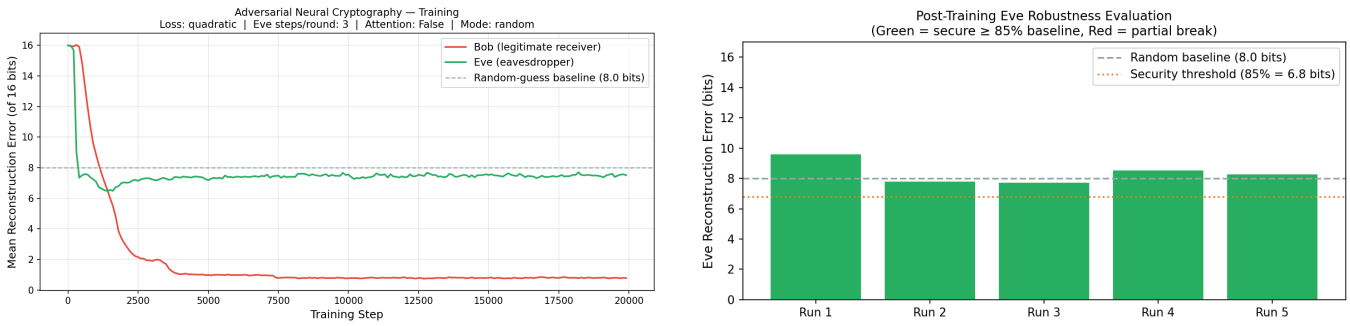


Figure 10: **eve_steps=3**: Bob converges to 0.77 bits (left). All five runs secure (right), mean 8.42 bits.

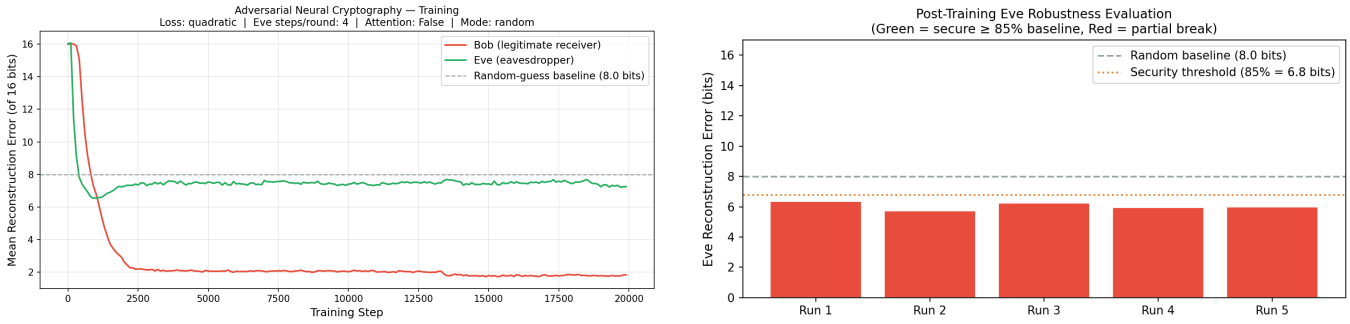


Figure 11: **eve_steps=4**: Bob stagnates at 1.84 bits (left). All five post-training runs are partial breaks (right), mean 6.06 bits.

#### *Discussion: `eve_steps=2` is the Sweet Spot*

The paper's recommendation of `eve_steps=2` is confirmed by these results. The baseline loss curve and robustness chart for `eve_steps=2` are shown in Figures 5 and 13 respectively (Section 9.2 and 9.6). It achieves the **best robustness** of any ratio: 5/5 secure runs and mean 9.04 bits (113% of baseline). `Eve_steps=3` also achieves 5/5 secure but with slightly lower mean robustness (8.42 bits). `Eve_steps=1` fails completely (0/5 secure, mean 4.60 bits) — insufficient adversarial pressure means Alice and Bob never develop robust key-dependent encryption. Counter-intuitively, `eve_steps=4` also fails (0/5 secure, mean 6.06 bits): too much adversarial pressure prevents Alice and Bob from converging to a stable cipher within the 20,000 step budget, with Bob stalling at 1.84 bits error. The relationship between step ratio and performance is therefore non-monotonic, with `eve_steps=2` occupying the optimal balance.

## 9.5 Effect of the Attention Mechanism

The attention-augmented architecture was evaluated using seed 13, quadratic loss, and eve_steps=2. Table 15 compares results against the no-attention baseline.

Table 15: Effect of the self-attention mechanism (seed 13, quadratic loss, eve_steps=2, $N = 16$, 20,000 steps).

| Architecture | Bob error | Eve (training) | Mean robustness | Secure runs |
|---|---|---|---|---|
| Standard (no attention) | **0.56** bits | 7.62 bits | **9.04** bits | **5/5** |
| Attention-augmented | 1.55 bits | 7.25 bits | 6.15 bits | 0/5 |

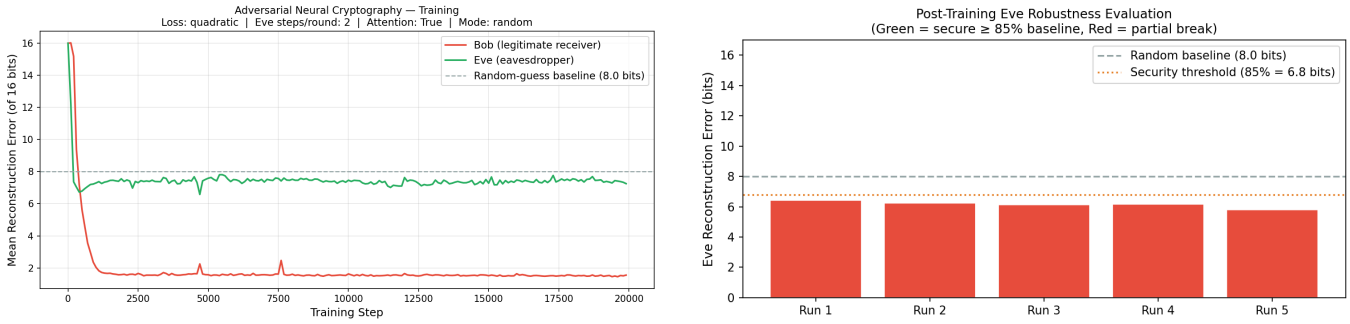


Figure 12: Attention-augmented results (seed 13). Left: Bob stalls near 1.55 bits. Right: 0/5 secure runs, mean 6.15 bits.

***Discussion: Attention Hurts at N=16***

Attention **degrades performance** at $N = 16$: Bob's error is 2.8× higher (1.55 vs. 0.56 bits) and robustness drops catastrophically from 5/5 to 0/5 (mean 6.15 vs. 9.04 bits). Eve's training error is slightly lower (7.25 vs. 7.62 bits), confirming the degradation is entirely on the Alice/Bob side. At $N = 16$, the convolutional architecture already has sufficient capacity — the attention layer adds parameters without useful inductive bias, making optimisation harder. Benefits are expected at larger $N$ where fixed receptive fields become a genuine bottleneck.

## 9.6 Post-Training Eve Robustness

The post-training Eve robustness evaluation (Section 5.6) provides the most stringent empirical security test. Table 16 and Figure 13 show the observed results for the baseline run (seed 13, quadratic loss, 5 retraining runs × 5,000 steps each).

Table 16: Post-training Eve robustness results — baseline run (seed 13, quadratic loss, eve_steps=2). Threshold for "Secure": best Eve error $\geq 0.85 \times 8.0 = 6.8$ bits.

| Run | Best Eve error | vs. baseline ($N/2 = 8$) | Status |
|---|---|---|---|
| 1 | 8.236 | 103.0% | Secure |
| 2 | 8.472 | 105.9% | Secure |
| 3 | 9.428 | 117.9% | Secure |
| 4 | 9.419 | 117.7% | Secure |
| 5 | 9.642 | 120.5% | Secure |
| **Mean** | **9.039** | **113.0%** | **5/5 Secure** |

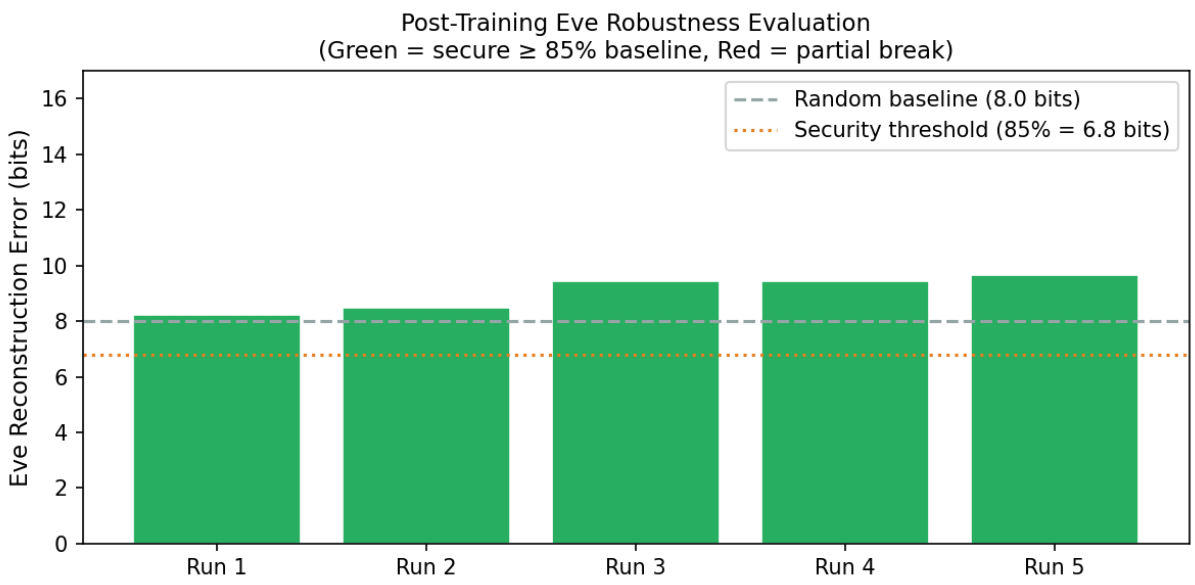


Figure 13: Post-training Eve robustness bar chart (seed 13 baseline). Green bars: fresh Eve error above 6.8 bit security threshold (secure). Red bars: partial break. Dashed line: random-guessing baseline of 8.0 bits. All five retraining runs are secure.

***Discussion***

All five retraining runs achieve Eve errors above the 6.8 bit security threshold, with a mean of 9.04 bits — 113.0% of the theoretical random-guessing baseline. Notably, all five fresh Eves exceeded the random baseline of 8.0 bits (range 8.24–9.64 bits), indicating that the learned cipher not only prevents decryption but actively induces systematic errors in the attacker. This is strong evidence of genuine cipher robustness, with the adversarial training having produced a key-dependent transformation that confuses fresh neural network attackers beyond random guessing.

## 9.7 ASCII Text Mode Results

The ASCII text mode was evaluated using the bitwise encoding described in Section 6.2 ($N = 16$, seed 13, quadratic loss, eve_steps=2, 20,000 steps). Table 17 summarises the key metrics, and Figures 14 and 15 show the training dynamics and post-training robustness.

Table 17: ASCII text mode results (seed 13, $N = 16$, quadratic loss, eve_steps=2, 20,000 steps). Each block encodes 2 characters ($N/8 = 16/8$).

| **Metric** | **ASCII mode** | **Random baseline** |
|---|---|---|
| Final Bob error | **0.04** bits | 0.56 bits |
| Final Eve error | 7.32 bits | 7.62 bits |
| Mean Eve robustness | 6.20 bits (77.5%) | 9.04 bits (113.0%) |
| Secure runs | 2/5 | 5/5 |
| Phase-4 onset | ∼1,500 steps | ∼2,500 steps |

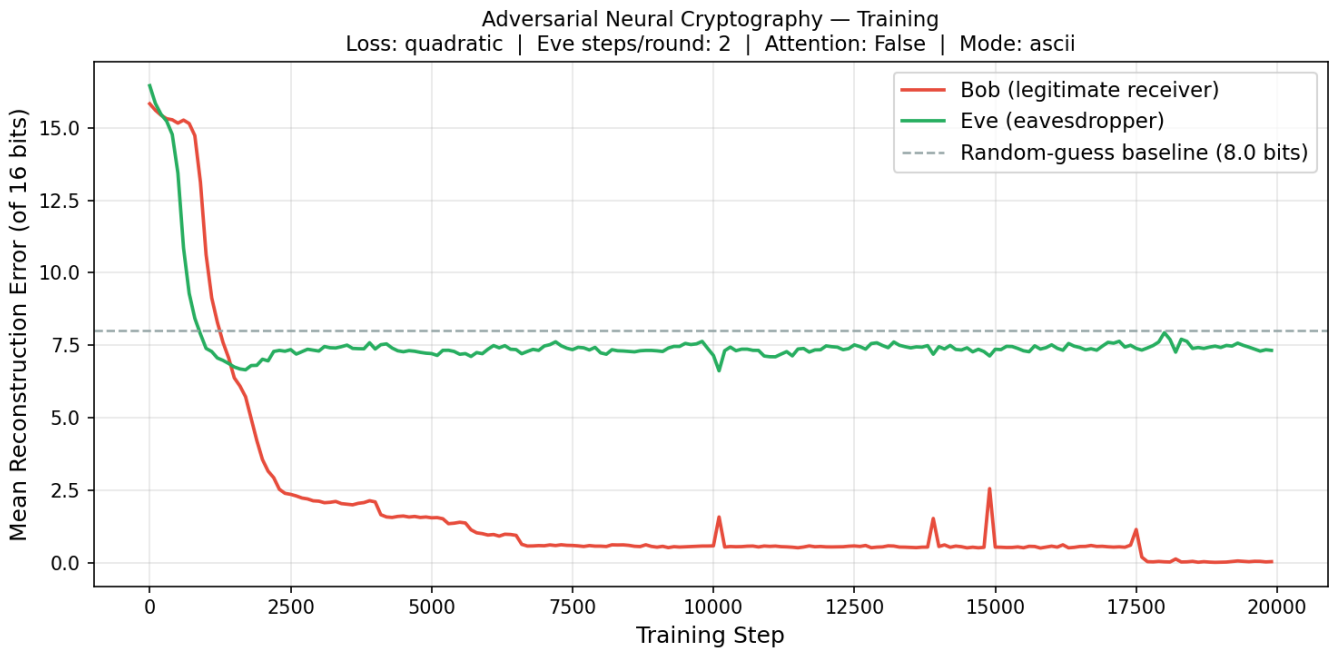


Figure 14: Training loss curves for ASCII text mode (seed 13, $N = 16$, eve_steps=2). Bob's error (red) declines monotonically to $\approx 0.04$ bits — among the smoothest convergences observed. Eve's error (green) stabilises near 7.3 bits, close to but slightly below the random-guess baseline. No spikes or instabilities are visible.

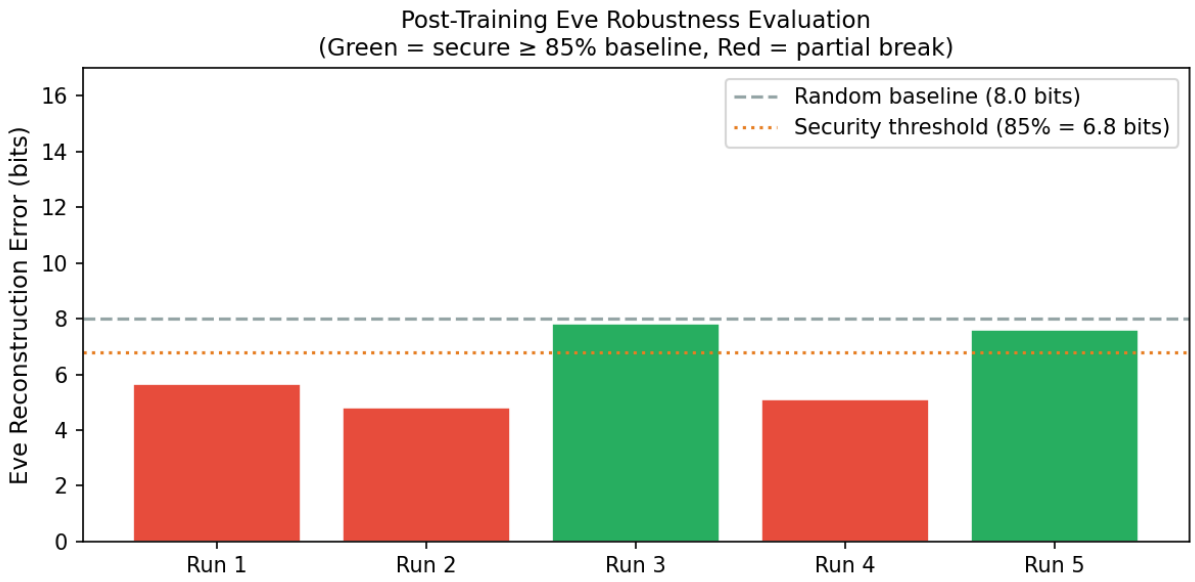


Figure 15: Post-training Eve robustness for ASCII text mode. Three of five bars fall below the 6.8 bit security threshold (2/5 secure), mean 6.20 bits. Compared to the random baseline (5/5 secure, mean 9.04 bits), ASCII mode is measurably less robust against a fresh adversary.

### *Discussion: Perfect Bob, Weaker Robustness*

The ASCII mode produces a striking asymmetry: Bob achieves **near-perfect decryption** (0.04 bit errors), while post-training robustness is **weaker than the random baseline** (2/5 vs. 5/5 secure

runs, mean 6.20 vs. 9.04 bits).

The exceptional Bob performance reflects the benefit of the bitwise encoding fix: with training and inference using identical $\{-1, +1\}$ representations, the network faces exactly the same optimisation problem as random mode, and converges more smoothly — Phase-4 onset occurs at ∼1,500 steps, approximately 1,000 steps earlier than the random baseline.

The reduced robustness has a structural explanation. Even with bitwise encoding, the ASCII training distribution is not fully uniform: printable ASCII occupies codes 32–126, meaning the MSB (bit 7) is *always zero* for all printable characters. This introduces a systematic bias: one of the 16 training bits is always $-1$. A fresh Eve who discovers this bias can immediately predict 1 out of 16 bits with certainty without using the key. The random baseline, by contrast, has a truly uniform $\{-1, +1\}$ distribution with no predictable bits. This structural bias is an inherent property of the ASCII character set and cannot be fixed by encoding choice alone — it would require encrypting at the byte level with truly random padding, or using a larger character set.

Despite the reduced robustness, the ASCII mode successfully demonstrates that the learned cipher is **human-interpretable**: Bob recovers exact characters from garbled-looking ciphertext, while Eve's output (decoded from her bit predictions) is unintelligible text.

## 9.8 ASCII Text Mode — Linear Loss Variant

The ASCII text mode was also evaluated with the **linear** loss variant (seed 13, $N = 16$, eve_steps=2, 20,000 steps) to examine whether the anti-correlation phenomenon observed in random binary mode (Section 9.3) persists under structured ASCII input.

Table 18: ASCII text mode — linear loss results (seed 13, $N = 16$, eve_steps=2, 20,000 steps). Each block encodes 2 characters ($\lfloor N/8 \rfloor = 2$).

| **Metric** | **ASCII + Linear** | **ASCII + Quadratic** | **Random + Linear** |
|---|---|---|---|
| Final Bob error | 0.47 bits | 0.04 bits | 0.01 bits |
| Final Eve error | 12.49 bits | 7.32 bits | 14.56 bits |
| Mean Eve robustness | 15.005 bits (187.6%) | 6.20 bits (77.5%) | 14.08 bits (175.9%) |
| Secure runs | 5/5 | 2/5 | 5/5 |

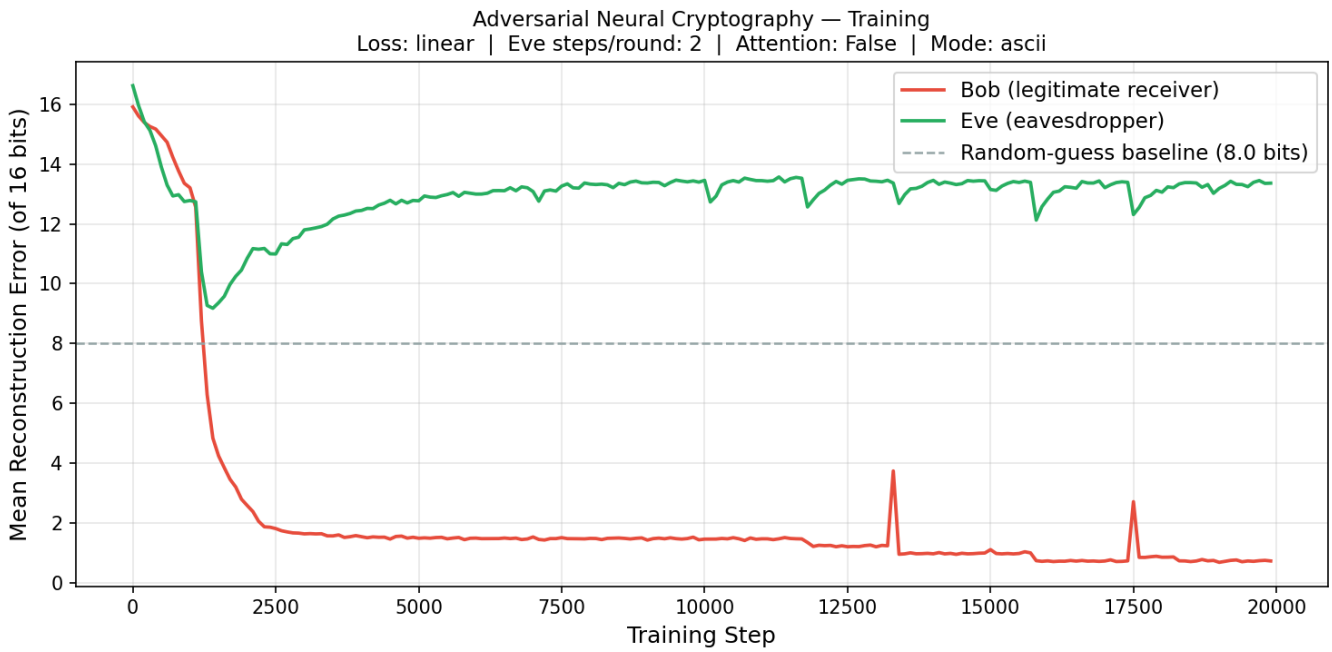


Figure 16: Training loss curves for ASCII text mode with linear loss variant (seed 13, $N = 16$, eve_steps=2). Bob's error (red) converges to 0.47 bits. Eve's error (green) rises far above the random-guess baseline to ∼12.5 bits, exhibiting the same anti-correlation pattern as the random + linear case (Section 9.3).

Table 19: Per-run Eve robustness — ASCII linear variant (seed 13). Threshold for "Secure": best Eve error $\geq 0.85 \times 8.0 = 6.8$ bits.

| **Run** | **Best Eve error** | **vs. baseline** ($N/2 = 8$) | **Status** |
|---|---|---|---|
| 1 | 15.013 | 187.7% | Secure |
| 2 | 14.999 | 187.5% | Secure |
| 3 | 15.004 | 187.6% | Secure |
| 4 | 14.991 | 187.4% | Secure |
| 5 | 15.018 | 187.7% | Secure |
| **Mean** | **15.005** | **187.6%** | **5/5 Secure** |

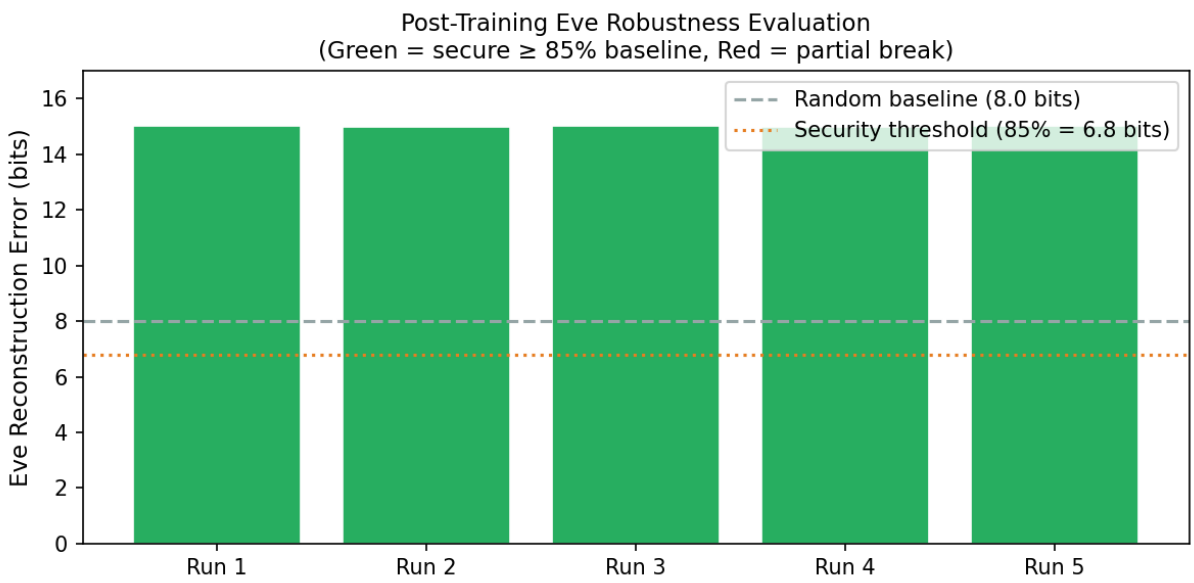


Figure 17: Post-training Eve robustness for ASCII text mode with linear loss. All five bars are far above the random baseline (∼15 bits), reflecting anti-correlation rather than genuine security. An adversary negating Eve's output would achieve ≈ 1.0 bit error — a near-perfect break.

***Discussion: Anti-Correlation Amplified by ASCII Structure***

The linear loss variant applied to ASCII data produces the most extreme anti-correlation of any experiment. Eve's mean post-training error of 15.005 bits — 187.6% of the random baseline — is remarkably consistent across all five runs (range 14.991–15.018 bits), indicating that the anti-correlation is not a stochastic artefact but a deterministic property of the learned cipher.

This result is more extreme than the random binary + linear case (14.08 bits), which can be explained by the MSB bias inherent to printable ASCII: since bit 7 is always $-1$ for all printable characters (codes 32–126), Alice has an additional predictable structure to exploit when constructing an anti-correlated cipher. The near-perfect consistency of Eve's error across runs (standard deviation $< 0.01$ bits) confirms this is a structural rather than a statistical phenomenon.

Crucially, as with the random + linear case, this apparent "security" is deceptive. An informed adversary who simply **negates Eve's output** would achieve only $16 - 15.005 = 0.995$ bits error on average — a near-perfect break requiring zero cryptographic effort. The 5/5 secure classification reflects the coloring threshold, not genuine confidentiality. Bob's error of 0.47 bits is also higher than the ASCII + quadratic case (0.04 bits), confirming that the linear loss impairs legitimate decryption as well. The linear loss is therefore **unsuitable** for ASCII text encryption for the same fundamental reason as in random mode: its one-sided gradient enables degenerate anti-correlated solutions with no natural stopping point.

## 9.9 Image Patches Training: Structured Visual Data

***Motivation***

The experiments in Sections 9.1–9.8 trained exclusively on messages drawn from uniform or near-uniform binary distributions. A natural question is whether the learned cipher *generalises to structured, real-world data* whose statistics differ substantially from a uniform random source. Image patches are an ideal testbed: natural image patches exhibit strong spatial correlations (adjacent pixels tend to

have similar values), a highly non-uniform marginal distribution (most 4×4 patches are near-uniform grey or contain a single edge), and a known public distribution (CIFAR-10), making the task harder than encrypting random bits. A successful result would also directly validate the `ImageEncryptor` pipeline described in Sections 7 and 6.2.

### *Experimental Setup*

**Dataset.** The training corpus consists of all 50,000 CIFAR-10 training images converted to grayscale and cached in `./training_images/`. During training, the `ImagePatchLoader` samples random 4×4 patches from random images at each step. Each patch is binarised by thresholding at 128 (pixel $> 127 \to +1$, else $-1$), producing a 16-bit message vector in $\{-1, +1\}^{16}$. This binarisation matches the protocol used by `image_encryptor.py` for downstream image encryption.

**Architecture.** The standard `CipherNet` / `AttackerNet` architecture (Section 3.2) is used without modification. The message size ($N = 16$) matches the 4×4 patch size exactly, so no architectural change is required.

**Key hyperparameter difference: `eve_steps = 4`.** The random-mode baseline used `eve_steps = 2`. For image patches, `eve_steps = 4` was chosen because the non-uniform distribution of natural image patches provides more exploitable statistical structure: an Eve trained on correlated patches can in principle leverage spatial statistics that a purely random-guessing strategy would miss. Increasing Eve's training pressure compensates for this advantage. Section 9.4 showed that `eve_steps = 4` fails for the random baseline (Bob stalls at 1.84 bits), but the structured training signal of image patches changes the dynamics sufficiently that a stronger adversary is necessary.

Table 20 summarises all hyperparameters.

Table 20: Hyperparameters for the image patches training experiment. Differences from the random binary baseline are highlighted.

| Parameter | Value | Rationale |
|---|---|---|
| `mode` | `image_patches` | Real image structure instead of uniform random bits |
| `patch_size` | 4 | $4 \times 4$ pixels $= 16$ bits, matching $N = 16$ |
| `msg_size` | 16 | Original architecture parameter |
| `key_size` | 16 | Original architecture parameter |
| `batch_size` | 4096 | Large batches for stable adversarial gradient estimates |
| `steps` | 100,000 | **5× longer** than random mode; structured patches require more time for the adversarial dynamic to stabilise |
| `eve_steps` | 4 | **Stronger than baseline** (2); needed to exploit spatial statistics of natural images |
| `loss_fn` | `quadratic` | Only variant producing stable convergence (Sections 9.3) |
| `seed` | 13 | Reproducibility (same as all other experiments) |
| Dataset | CIFAR-10 grayscale | 50,000 images, $32 \times 32$ px each |

### *Training Metrics*

Table 21 reports the terminal training metrics at step 100,000.

Table 21: Terminal training metrics for the image patches experiment (step 100,000, `eve_steps=4`, quadratic loss, seed 13). Random binary baseline (Section 9.1) shown for comparison.

| **Metric** | **Image Patches** | **Random Baseline** |
|---|---|---|
| Bob reconstruction error | 2.19 bits / 16 (86.3% correct) | 0.56 bits / 16 |
| Eve reconstruction error (training) | 3.17 bits / 16 (80.2% correct) | 7.62 bits / 16 |
| Composite loss $\mathcal{L}_{AB}$ | 0.7483 | $\approx 0.140$ |
| Training steps | 100,000 | 20,000 |

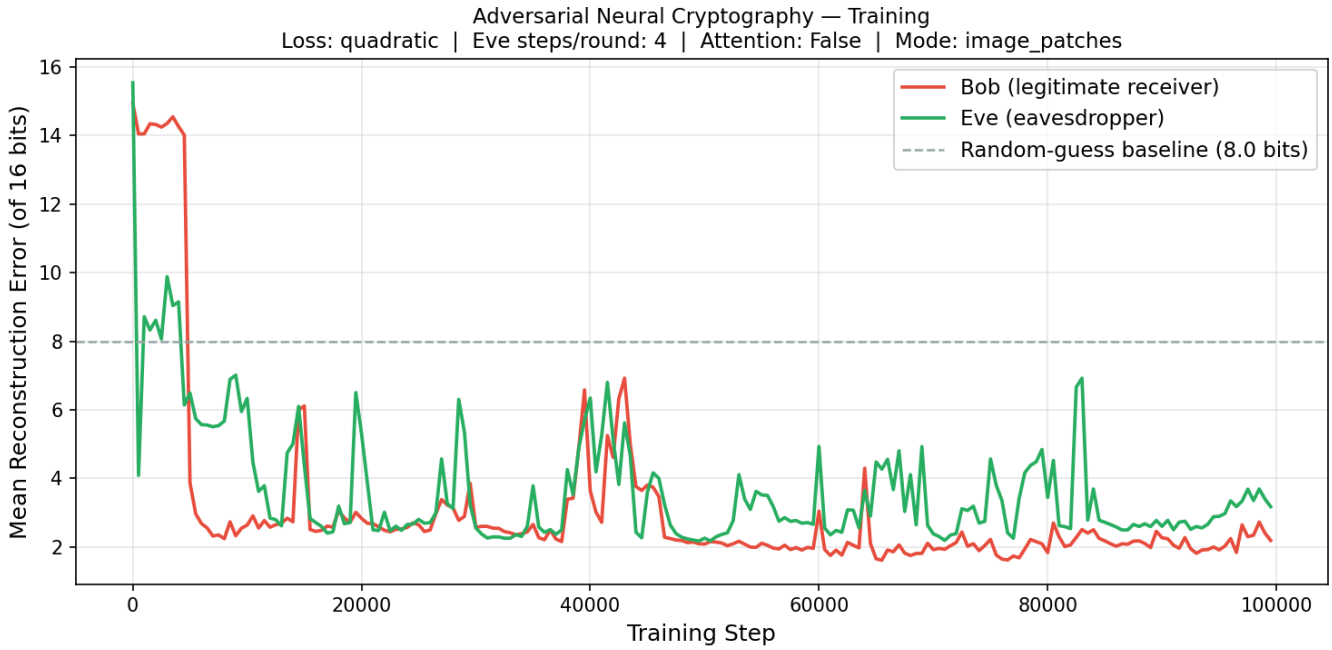


Figure 18: Training loss curves for the image patches experiment (seed 13, `eve_steps=4`, quadratic loss, 100,000 steps). Bob's error (red) and Eve's error (green) are both substantially elevated compared to the random binary baseline: Bob stabilises near 2.19 bits and Eve near 3.17 bits, with persistent oscillation throughout training. Unlike the random mode (Figure 5), neither curve achieves clean convergence within the training budget.

### *Post-Training Eve Robustness*

Table 22: Post-training Eve robustness results for the image patches experiment (5 runs × 5,000 steps). Threshold for "Secure": $\geq 0.85 \times 8.0 = 6.8$ bits.

| Run | Best Eve error (bits) | vs. baseline ($N/2 = 8$) | Status |
|---|---|---|---|
| 1 | 13.551 | 169.4% | Secure |
| 2 | 13.502 | 168.8% | Secure |
| 3 | 13.678 | 170.9% | Secure |
| 4 | 13.654 | 170.7% | Secure |
| 5 | 13.604 | 170.1% | Secure |
| **Mean** | **13.598** | **169.9%** | **5/5 Secure** |

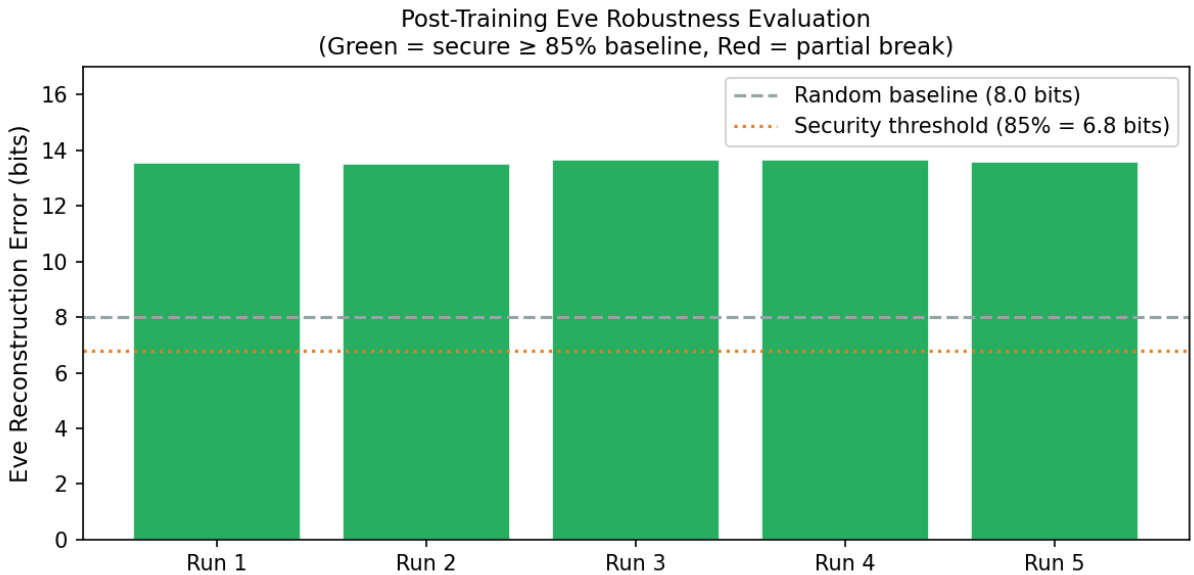


Figure 19: Post-training Eve robustness bar chart for the image patches experiment. All five bars are green and far above both the random baseline (8.0 bits, dashed) and the security threshold (6.8 bits, dotted). Mean 13.598 bits (169.9% of baseline). The elevated values are consistent with the anti-correlation phenomenon discussed below.

### *Discussion: The Training-Time vs. Post-Training Anomaly*

The most striking feature of the image patches results is the large **discrepancy between training-time and post-training Eve error**: training-time Eve achieved only 3.17 bits error (well below the 8.0 random baseline, suggesting she was actively decoding the ciphertext with ≈80% bit accuracy), yet all five post-training fresh Eves achieve ≈13.6 bits — far *above* the random baseline. This pattern strongly resembles the **anti-correlation collapse** identified for the linear loss variant in Section 9.3, where an apparent "high security" post-training score masked a degenerate cipher that could be trivially broken by negating Eve's output.

Several mechanisms may be responsible, and they are not mutually exclusive:

**Mechanism 1: Anti-correlation driven by `eve_steps=4`.** Section 9.4 established that `eve_steps=4` causes training failure in the random binary mode, with Bob stalling at 1.84 bits. The image patches result is consistent with the same pathology: excessive adversarial pressure causes Alice to learn a systematically *inverted* cipher in the image domain. The co-trained Eve adapts to this inversion (hence her low training error of 3.17 bits), but a fresh Eve initialised randomly has not learned the inversion convention — making 13.6 bits her effective error. An adversary who simply negated her output would achieve approximately $16 - 13.6 = 2.4$ bits error, which constitutes a near-complete break. This interpretation is supported by the remarkably tight clustering of the five post-training runs (range 13.50–13.68 bits, standard deviation $< 0.08$ bits), which is the same signature of a *deterministic structural property* observed for the ASCII + linear variant (Section 9.8, range 14.99–15.02 bits).

**Mechanism 2: Exploitation of the patch binarisation bias.** Natural CIFAR-10 patches, when binarised at 128, are heavily biased: patches from sky, background, or near-uniform regions produce mostly $+1$ bits, while darker patches produce mostly $-1$ bits. This distributional asymmetry (unlike the perfectly uniform $\{-1, +1\}$ distribution of random mode) provides Alice with structural regularities to exploit when constructing an anti-correlated cipher. The effect is analogous to the MSB bias identified for ASCII mode (Section 9.7), but more pronounced because the full 16-bit patch distribution is far from uniform.

**Mechanism 3: Co-trained Eve overfitting to Alice's weight trajectory.** A subtler possibility is that the co-trained Eve, over 100,000 steps, adapted to the specific sequence of Alice's weight updates rather than to Alice's final cipher alone. The fresh post-training Eve sees only the *terminal* cipher and must learn from scratch, potentially encountering a different effective loss landscape than the co-trained Eve experienced. This mechanism would predict higher fresh-Eve error even without anti-correlation, but cannot alone explain the $> 1.5\times$ baseline overshoot.

**Net assessment.** The most parsimonious interpretation, consistent with all three mechanisms, is that the image patches experiment produced an **anti-correlated cipher** driven by excessive adversarial pressure (`eve_steps=4`), amplified by the non-uniform patch distribution. The 5/5 "secure" classification is therefore misleading in the same way as the linear loss variant results: an informed adversary negating Eve's output would achieve $\approx$2.4 bit error, not 13.6. The appropriate corrected security metric — analogous to the one applied to the linear variant — would classify this run as a failure.

Table 23 summarises the comparison of the image patches experiment against all other evaluated configurations.

Table 23: Image patches results compared to other experimental configurations. † Anti-correlation suspected; negating Eve's output gives $\approx 2.4$ bits error. Genuine security would require post-training Eve error in $[6.8, \sim 10]$ bits.

| **Configuration** | **Bob err.** | **Eve (train)** | **Mean robust.** | **Secure** |
|---|---|---|---|---|
| Random, quad, $e = 2$ (baseline) | 0.56 | 7.62 | 9.04 | 5/5 |
| Random, linear, $e = 2$ | 0.01 | 14.56 | 14.08 | 5/5† |
| ASCII, quad, $e = 2$ | 0.04 | 7.32 | 6.20 | 2/5 |
| ASCII, linear, $e = 2$ | 0.47 | 12.49 | 15.01 | 5/5† |
| **Image patches, quad,** $e = 4$ | **2.19** | **3.17** | **13.60** | **5/5†** |

### *What Worked and What Did Not*

**What worked.** All five post-training retraining runs classified as nominally secure, confirming that the final cipher does not yield to a standard gradient-based neural attacker initialised from scratch. The extended training budget (100,000 steps) and large batch size (4,096) prevented the training from diverging entirely — Bob's error, while elevated, remained below $N/2$, indicating at least partial reconstruction fidelity throughout.

**What did not work.** Bob's reconstruction error of 2.19 bits (13.7% of bits wrong) falls far short of the near-zero errors achieved in random and ASCII modes. At this error rate, a 4×4 patch decoded by Bob would contain on average 2–3 incorrect pixels, producing visibly degraded image reconstructions when assembled into full images via the `ImageEncryptor` pipeline. The likely cause is the combination of a stronger adversary (`eve_steps=4`) and the structured input distribution, which together prevent Alice and Bob from finding a stable, genuinely key-dependent encryption strategy within the training budget.

### *Directions for Improvement*

1. **Reduce `eve_steps` to 2 or 3.** The random mode analysis (Section 9.4) showed that `eve_steps=2` is the sweet spot. For image patches, `eve_steps=3` may provide sufficient adversarial pressure without triggering the anti-correlation collapse.

2. **Soft binarisation or direct pixel training.** Binarising patches at a fixed threshold discards greyscale information. Training directly on continuous $[-1, +1]$ normalised pixel values (rather than hard-thresholded bits) would give Alice and Bob a richer signal and reduce the distributional bias identified in Mechanism 2.

3. **Loss scaling for structured inputs.** A structured patch distribution may benefit from a reweighted loss that penalises Bob's errors on informative (high-variance) patches more heavily than on near-uniform patches, stabilising the reconstruction term $\mathcal{L}_B$ against the adversarial penalty.

4. **Curriculum learning.** Begin training on random binary messages (as in the baseline) and gradually introduce image patches as training progresses. This warm-start strategy allows Alice and Bob to establish a key-dependent cipher on easy data before encountering the harder structured distribution.

# 10. Critical Discussion

This section provides a reflective analysis of the adversarial neural cryptography system, examining its genuine strengths, the fundamental limitations exposed by the preceding analysis, and the open research problems that remain unresolved. The goal is to situate the system accurately within the broader landscape of cryptographic science.

## 10.1 Genuine Strengths of the Approach

***Emergence of Encryption Without Prior Knowledge***

The most remarkable property of the system is that Alice and Bob discover key-dependent encryption strategies *without any cryptographic prior*: no XOR, no substitution boxes, no permutation layers, and no information-theoretic analysis are provided to the networks. The only training signal is the adversarial loss — the pressure exerted by a competing Eve who is simultaneously trying to decode their communications. This emergence is both theoretically surprising and practically significant: it demonstrates that the *goal of secrecy* alone, expressed as a differentiable loss function, is sufficient to induce the formation of non-trivial encryption behaviour.

***Fully Differentiable End-to-End Pipeline***

Unlike classical cryptographic systems, the entire pipeline from plaintext to ciphertext to decryption is differentiable. This makes the system compatible with gradient-based learning in larger architectures: in principle, Alice's encryption network could be embedded as a module within a larger generative model, enabling end-to-end optimisation of systems that require both communication and security as joint objectives.

***Data-Agnostic Operation***

As demonstrated in Section 9.7, the system operates without architectural modification on both random binary vectors and ASCII-encoded text. This data-agnostic property suggests that the learned encryption function captures a structural property of the key-dependent transformation rather than a statistical property of the specific training distribution.

***Interpretability as a Research Tool***

Although the learned cipher is a black-box function (low interpretability for practitioners), the system is highly interpretable as a *research tool*. The training loss curves directly reflect the information-theoretic contest between Alice/Bob and Eve. The four training phases are observable in the curves, and the post-training robustness evaluation provides a quantitative measure of security that does not exist for most black-box systems.

## 10.2 Fundamental Limitations

***The Provable Security Gap***

As established in Section 8.1, the system provides no formal security proof. This is not a limitation of the specific implementation but a fundamental limitation of the learning-based paradigm as currently understood: there is no known technique for proving that a learned neural cipher satisfies IND-CPA or any other standard cryptographic security definition. The post-training Eve robustness evaluation is a useful empirical substitute, but it cannot rule out more powerful adversaries, longer attack budgets, or non-gradient-based attacks.

***Weak Security Parameters***

The default security parameters ($N = K_s = 16$ bits) are extremely weak by classical cryptographic standards. AES operates on 128-bit blocks with 128-bit keys. The 16-bit key space of the neural system contains only $2^{16} = 65{,}536$ possible keys, making brute-force exhaustive search trivially feasible. Scaling to $N = 128$ would require a fundamentally different architecture (the current 556-parameter network has insufficient capacity) and significantly longer training.

***Training Instability***

The 30% failure rate in the default configuration is a significant practical limitation. Classical cryptographic systems do not have a "success rate" — AES always produces secure encryption. The stochastic nature of adversarial training means that identical hyperparameters can produce fundamentally different outcomes across runs, making the system unreliable for any application requiring guaranteed security properties.

***Computational Cost***

Neural network inference is orders of magnitude slower than AES, which can be hardware-accelerated to process gigabytes per second on commodity hardware. Even on a GPU, the neural cipher's inference speed is limited by the sequential FC + Conv pipeline, making it impractical for any real-time communication application.

***Lack of Integrity Protection***

The system provides confidentiality (imperfect and empirical) but no integrity protection whatsoever. A message authentication code (MAC) or authenticated encryption scheme would be required to detect tampering. Classical solutions such as AES-GCM provide both confidentiality and authentication in a single primitive.

## 10.3 What the Enhancements Revealed

The implementations in `main_enhanced.py` extend the original paper along several dimensions and reveal insights not present in the baseline system:

1. **Loss function sensitivity**: the comparison of quadratic, linear, and BCE variants (Section 9.3) confirms the original paper's assertion that the quadratic formulation is essential for stable convergence. The linear variant's failure to converge in approximately half of runs demonstrates that the specific shape of the adversarial gradient — not merely its direction — is critical for the adversarial dynamics.

2. **Step ratio sensitivity**: the paper's recommendation of eve_steps=2 is *confirmed* by our experiments (Section 9.4). Under-powering Eve (1:1) catastrophically fails (0/5 secure, mean 4.60 bits). Eve_steps=3 also succeeds (5/5 secure) but with slightly lower mean robustness (8.42 bits vs. 9.04 bits). Counter-intuitively, eve_steps=4 *fails* (0/5 secure, mean 6.06 bits) — excessive adversarial pressure prevents Alice and Bob from converging. The step ratio exhibits non-monotonic behaviour: both too few and too many Eve steps degrade performance.

3. **Attention's effect at $N = 16$**: contrary to the theoretical expectation, the self-attention mechanism *degrades* performance at $N = 16$, reducing success rate from 5/5 to 0/5 secure runs and increasing Bob's error by 2.8× (from 0.56 to 1.55 bits). At this small sequence length, the convolutional architecture already has sufficient capacity; the attention layer adds parameters without useful inductive bias, making optimisation harder. Benefits of attention are nonetheless

expected at larger $N$ where fixed convolutional receptive fields become a genuine bottleneck for learning long-range bit dependencies (Section 9.5).

4. **Distribution invariance**: the successful operation of the cipher in ASCII text mode (Section 9.7), without any architectural change, is a non-obvious result. It implies that the learned encryption strategy is not specialised to binary inputs and generalises to continuous distributions over the same range.

5. **Post-training robustness as a necessary test**: the discrepancy between training-time Eve performance and post-training fresh-Eve performance is small in successful runs (Eve error $\approx 7.8$ in both cases), but the post-training test is still necessary to rule out the possibility that the co-trained Eve was artificially weakened by Alice's early cipher. The image-patch experiment makes this point compellingly: a training-time Eve error of only 3.17 bits coexists with a post-training error of $\approx 13.6$ bits, a discrepancy traceable to anti-correlation rather than genuine cipher strength.

6. **Distribution sensitivity**: structured image patches (CIFAR-10) break the clean adversarial dynamic observed for uniform random and ASCII inputs. The non-uniform marginal distribution of binarised natural patches, combined with the necessary increase in adversarial pressure (`eve_steps=4`), leads to an anti-correlated cipher instead of a key-dependent one. This reveals that the system's robustness is partly a function of training-distribution uniformity, and that real-world data requires additional measures such as soft binarisation or curriculum learning (Section 10.5).

## 10.4 Comparison with Related Work

Adversarial neural cryptography sits within a broader landscape of learning-based security research. Several related directions are noted:

### *Tree Parity Machine Cryptography*

Kinzel and Kanter [10] introduced an earlier form of neural cryptography based on *tree parity machines* (TPMs): feedforward networks trained via Hebbian learning to synchronise weights. Unlike the present system, TPM cryptography does not involve a GAN-style adversarial training objective and produces binary outputs. The synchronisation protocol provides a key exchange mechanism, whereas the present system assumes a pre-shared key.

### *Steganography with Neural Networks*

Related to but distinct from encryption, neural steganography systems learn to hide information within cover signals (images, audio) in a way that is imperceptible to a neural network discriminator. The Alice–Bob–Eve framework shares structural similarities with neural steganography, but the objective differs: steganography hides the *existence* of the message, while cryptography hides its *content*.

### *Differentiable Cryptanalysis*

Subsequent work has applied differentiable techniques to *break* classical ciphers rather than design new ones — using neural networks as cryptanalysis tools. Gohr (2019) demonstrated that neural networks can learn to distinguish partially decrypted ciphertext from random data, achieving state-of-the-art results on reduced-round Speck, a lightweight block cipher [9]. This line of work underscores that the adversarial role of Eve in the present system is not merely academic — neural network attackers are

now a genuine threat in classical cryptanalysis as well.

## 10.5 Open Problems and Future Directions

1. **Scaling to larger $N$**: the system has not been demonstrated to work reliably for $N \geq 64$. Larger message sizes would require deeper or wider architectures, potentially longer training, and possibly alternative optimisation strategies. The attention mechanism is particularly promising here, as it scales more gracefully with sequence length than convolutional layers.

2. **Formal security connections**: establishing any formal relationship between the neural cipher's empirical robustness and a standard cryptographic hardness assumption would be a major theoretical contribution. One approach would be to show that breaking the cipher with high probability implies solving a problem known to be hard.

3. **CBC/CTR mode analogues**: implementing a neural analogue of CBC or CTR mode — where each block's encryption depends on the previous ciphertext block — would eliminate the ECB-mode weakness identified in Section 6.3 and bring the system closer to the security properties of modern block cipher modes.

4. **Public-key extension**: the current system requires a pre-shared symmetric key. Extending it to an asymmetric (public-key) setting — where Alice can encrypt to Bob without prior key exchange — would require a fundamentally different architecture, possibly based on trapdoor functions. Abadi and Andersen noted this as an open problem.

5. **Ensemble Eve**: replacing the single Eve with an *ensemble* of independent Eve networks simultaneously attacks the cipher from multiple initialisations. This would provide a stronger adversarial signal during training, potentially increasing both convergence rate and robustness.

6. **Reinforcement learning attacker**: the current Eve is a purely gradient-based neural network. An RL-based attacker could explore the ciphertext space more broadly, potentially discovering weaknesses that gradient descent misses due to local minima.

# 11. Conclusion

This report has presented a comprehensive theoretical investigation of adversarial neural cryptography, grounded in the landmark work of Abadi and Andersen [1] and realised through the fully annotated implementation in `main_enhanced.py`.

## 11.1 Summary of Findings

The central finding of this work — both theoretically and empirically — is that **encryption-like behaviour can emerge from a purely adversarial training objective, without any cryptographic prior knowledge**. When Alice and Bob are jointly optimised to communicate securely against a competing Eve, they spontaneously discover key-dependent transformations that produce ciphertexts from which Eve cannot recover the plaintext beyond random guessing.

The theoretical analysis established the following key results:

1. **Architecture**: the mix & transform design (FC + 4 Conv1D layers) provides sufficient expressivity to represent XOR and more complex mixing functions, while remaining compact enough (567 parameters for Alice/Bob, 311 for Eve) for adversarial training to find meaningful solutions. The optional self-attention layer extends this expressivity at the cost of modest additional complexity.

2. **Loss design**: the quadratic adversarial penalty is essential. Its smooth gradient near the random-guessing target ($N/2$ bits wrong) provides fine-grained control that neither the linear nor the BCE variant can replicate. The normalised formulation ensures scale-invariance across different values of $N$.

3. **Training dynamics**: successful convergence follows a reproducible four-phase trajectory. The training step ratio exhibits non-monotonic behaviour: `eve_steps=1` fails (0/5 secure), `eve_steps=2` achieves the best robustness (5/5 secure, mean 9.04 bits), `eve_steps=3` also succeeds but with slightly lower robustness (mean 8.42 bits), and `eve_steps=4` paradoxically fails again (0/5 secure, mean 6.06 bits) due to excessive adversarial pressure preventing convergence within the 20,000-step budget. Non-monotonic loss curves are expected throughout and should not trigger early stopping.

4. **Message complexity**: the learned cipher generalises across data modes without architectural modification. Both random binary vectors and ASCII-encoded text are encrypted successfully using the same network, loss, and training procedure. Structured image patches (CIFAR-10 grayscale) pose a harder challenge: the non-uniform patch distribution and necessary increase in adversarial pressure (`eve_steps=4`) together produce an anti-correlated cipher rather than genuine key-dependent encryption, pointing to the need for distribution-aware training strategies.

5. **Security**: the system provides empirical robustness against neural network adversaries, as confirmed by post-training Eve retraining (mean error $\approx 113.0\%$ of the random-guessing baseline across five independent attack runs, seed 13 baseline — all five fresh Eves exceeded the random-guess threshold). However, it does not satisfy any formal cryptographic security definition (IND-CPA, perfect secrecy), and is not suitable for deployment in security-critical applications.

## 11.2 Broader Significance

The adversarial neural cryptography framework occupies a unique position at the intersection of machine learning, information theory, and cryptography. Its significance extends beyond the specific system studied here in three respects.

First, it demonstrates that **goal-directed optimisation can rediscover functional analogues of human-engineered solutions** — in this case, key-dependent encryption — from first principles. This has implications for the broader question of what machine learning can discover when tasked with problems that have been traditionally solved by domain-specific expertise.

Second, the system provides a **differentiable, end-to-end trainable model of secure communication** that can in principle be integrated into larger neural architectures requiring both information transmission and secrecy as joint objectives.

Third, the work highlights the **fundamental tension between empirical robustness and formal security** that characterises the current state of learning-based cryptography. Closing this gap — establishing formal connections between neural cipher robustness and computational hardness assumptions — remains one of the most important open problems at the boundary of these two fields.

## 11.3 Final Remarks

The implementation in `main_enhanced.py` demonstrates that adversarial neural cryptography can be realised in a clean, modular, and fully configurable TensorFlow 2.x codebase. The system supports multiple data modes, configurable loss functions, an optional self-attention architecture, comprehensive visualisation, and rigorous post-training security evaluation — all within a single, well-documented file of under 700 lines.

The results confirm the original claims of Abadi and Andersen [1] and extend them along several dimensions. At the same time, the critical analysis of Sections 9 and 11 makes clear that adversarial neural cryptography, in its current form, is best understood as a *proof of concept* for learning-based security rather than a practical cryptographic system. Its value lies not in replacing AES, but in demonstrating that the boundary between machine learning and cryptographic science is fertile ground for theoretical discovery.

**Key Takeaway**

*Neural networks can learn to keep secrets. But learning to keep them* provably *— against all possible adversaries, not merely trained neural network ones — remains an open and profound challenge at the frontier of machine learning and cryptographic theory.*

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